

OmniTIEFormer: A tri-branch transformer with cross-scale transfer learning for multi-scale battery life-cycle forecasting

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HIGHLIGHTS

- Tri-branch Transformer captures local, regional, and global degradation cues.
- TCEM enables cross-scale feature exchange and gated fusion for forecasting.
- Outperforms SOTA: −44% MAE and −74% EOL error across multiple datasets.
- Robust on CALCE, PANASONIC, and GOTION, handling non-monotonic patterns.
- Transfers from small cells to large industrial batteries with strong accuracy.

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ABSTRACT

Accurately predicting the remaining useful life (RUL) of lithium-ion batteries is fundamentally challenging due to their multi-scale degradation dynamics, which range from approximately linear trends to complex nonlinear behaviors. Moreover, existing models often struggle to generalize across different battery types, limiting their practicality in real-world applications. To address these issues, we propose OmniTIEFormer, a novel multi-scale Transformer architecture designed for end-to-end (E2E) battery lifecycle management. OmniTIEFormer features a three-branch parallel architecture that simultaneously captures local fluctuations, regional trends, and global degradation patterns from historical capacity sequences. These multi-scale representations are fused by a Tri-branch Cross-Exchange Module (TCEM), which enables effective interaction and integration of information across scales. We demonstrate the effectiveness of OmniTIEFormer through extensive experiments on multiple public datasets exhibiting diverse degradation behaviors. OmniTIEFormer outperforms state-of-the-art baselines, reducing mean absolute error by up to 44% and end-of-life (EOL) prediction error by up to 74%. The model's robustness and generalization capability are further demonstrated in a challenging cross-dataset, cross-scale transfer learning setting: when pre-trained on small-capacity cells, it adapts effectively to large-format industrial batteries. This indicates that OmniTIEFormer captures fundamental, transferable degradation patterns rather than overfitting to specific datasets. These findings position OmniTIEFormer as a robust, data-efficient solution for battery prognostics, particularly valuable in industrial scenarios where data for new battery types is often limited. The code is publicly available at: <https://github.com/keepawakeyi/OmniTIEFormer>.

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1. Introduction

The global electrification transition in the transportation and energy sectors has propelled lithium-ion batteries to the forefront of modern energy systems. Effective management across the entire battery lifecycle, spanning design, operation, second-life utilization, and recycling, is essential for unlocking their full economic and environmental value. Within this lifecycle, accurate prediction of RUL is critical for ensuring system safety, enabling predictive maintenance, and objectively evaluating second-life potential [1–3]. However, battery degradation is far from a simple linear process. Instead, it is governed by complex non-linear dynamics operating across multiple temporal scales. The health trajectory typically exhibits a long-term macroscopic degradation trend intertwined with local capacity regeneration phenomena induced by rest periods [4,5], as well as occasional abrupt performance drops. Capturing these interacting patterns remains a fundamental challenge for existing prognostic models.

From physics-based models, whose generalization is often limited by parameter identification challenges [6,7], to modern data-driven approaches, the evolution of battery prognostics has consistently faced a central dilemma: how to model multi-scale degradation dynamics effectively. Early recurrent architectures such as LSTMs and GRUs [8] rely on sequential single-path processing, which often entangles short-term fluctuations and long-term degradation signals within a single hidden state. Hybrid approaches, such as the GRU–Kalman filter framework proposed by Song et al. [9], partially alleviate this limitation but do not fully resolve the underlying issue. The emergence of Transformer architectures [10–12] has significantly improved predictive performance; however, most implementations still follow a serial “extract-then-aggregate” paradigm. For instance, the Pre-trained Predictive Transformer (PPT) model [13] combines convolutional feature extraction with probabilistic sparse attention to capture both short- and long-term battery behaviors. Nevertheless, its serial architecture restricts bidirectional information flow across temporal scales, limiting the model’s ability to capture interactions between local events and global degradation trends. This limitation becomes particularly evident under domain shift: models trained on small laboratory cells often struggle when transferred to industrial batteries operating under more complex conditions and higher noise levels, where genuine physicochemical signals are more difficult to distinguish from transient disturbances.

Parallel architectures face similar challenges. For example, the TOP-Net model proposed by Chen et al. [14] employs a 2D–1D parallel hybrid neural network to learn multi-scale representations under diverse cycling conditions. However, because information exchange occurs only at the final fusion stage through concatenation, feature extraction in each branch remains largely isolated. Consequently, local branches that capture cycle-to-cycle fluctuations cannot effectively leverage global degradation trends to suppress noise, while global branches may overlook local evidence of capacity regeneration. As a result, both serial (e.g., PPT [13]) and parallel (e.g., TOP-Net [14]) architectures share a common bottleneck: cross-scale information interaction is confined to a single post-hoc fusion stage rather than being continuously performed through bidirectional exchange during representation learning. This limitation is further reflected in later experiments on datasets characterized by pronounced capacity regeneration and non-monotonic degradation.

Despite recent progress, two key limitations remain insufficiently addressed in existing battery prognostics models. First, most current architectures restrict cross-scale interaction to unidirectional or post-hoc fusion, limiting their ability to capture the dynamic interplay among local fluctuations, regional transitions, and global degradation trends. Second, the learned representations often generalize poorly across heterogeneous battery domains, reducing model robustness when battery chemistry, capacity scale, or operating conditions vary.

Addressing these challenges requires an architecture that supports continuous cross-scale information exchange during feature learning, rather than relying solely on post-hoc fusion. To this end, this

paper introduces the Omni-scale Tri-branch Information Exchange Transformer (OmniTIEFormer), a multi-scale predictive architecture designed to advance battery prognostics from weakly coupled multi-scale modeling toward concurrent and interactive representation learning. As shown in Fig. 1, we further outline an end-to-end (E2E) workflow for battery lifecycle management, in which OmniTIEFormer serves as the prognostic core. This workflow spans data acquisition, cloud-based analysis, prognostic inference, and subsequent management-oriented stages. Within this workflow, OmniTIEFormer is used to predict the future capacity trajectory, from which the EOL point and the RUL are subsequently estimated. Combined with sensor monitoring information, these prognostic results can inform subsequent decisions in Battery Management Systems (BMS), such as battery retirement, maintenance, second-life utilization, or recycling. This workflow illustrates how the proposed method can support battery lifecycle management tasks. Its tri-branch architecture is designed to capture complementary temporal scopes in battery degradation, thereby supporting both predictive accuracy and cross-domain generalization. To rigorously evaluate its robustness under realistic conditions, we conduct experiments on challenging datasets, including large-format industrial batteries (e.g., the 27 Ah GOTION dataset [15]) and the PANASONIC dataset [15], which exhibit significant capacity regeneration and non-monotonic degradation behavior.

The main contributions of this work are summarized as follows:

- 1. Methodological Contribution:** We propose OmniTIEFormer, a novel multi-scale predictive architecture for battery capacity prognostics. By integrating a tri-branch parallel design with the Tri-Branch Cross-Exchange Module (TCEM) and Local Hybrid Fusion Attention (L-HFA), the model enables concurrent modeling and bidirectional interaction of local, regional, and global degradation dynamics directly from historical capacity sequences.
- 2. A Prognostics-Centered Workflow for Lifecycle Management:** We propose a prognostics-centered conceptual workflow for battery lifecycle management and position our model as its core prognostic engine, aiming to bridge the gap between algorithmic research and practical industrial workflows.
- 3. Empirical Contribution:** We conduct comprehensive evaluations across multiple public datasets encompassing heterogeneous battery chemistries, capacities, and complex degradation patterns, including non-monotonic regeneration. The results demonstrate that OmniTIEFormer achieves robust predictive performance across diverse operating regimes. Compared with competitive baselines, it reduces the EOL prediction error by up to 74%. It exhibits strong transferability from laboratory-scale cells to large-format industrial batteries, confirming its potential in data-scarce and domain-shift scenarios.

1.1. Organization

The paper is organized as follows: Section 2 reviews related work. Section 3 presents OmniTIEFormer and its key modules. Section 4 describes the experimental setup and datasets. Section 5 reports results and analyses. Section 6 concludes the paper.

2. Related work

Methodologies for battery RUL prediction have evolved from first-principles models to the now-dominant data-driven paradigm [16–20]. Early data-driven studies relied heavily on feature engineering. For example, Severson et al. [21] showed that features extracted from the variance of early-cycle discharge curves can predict full cycle life. Such approaches, however, implicitly require sufficiently large and diverse cell populations to validate and generalize handcrafted descriptors; with limited samples, models may overfit and capture spurious correlations. This dependence on domain expertise and data is also reflected in

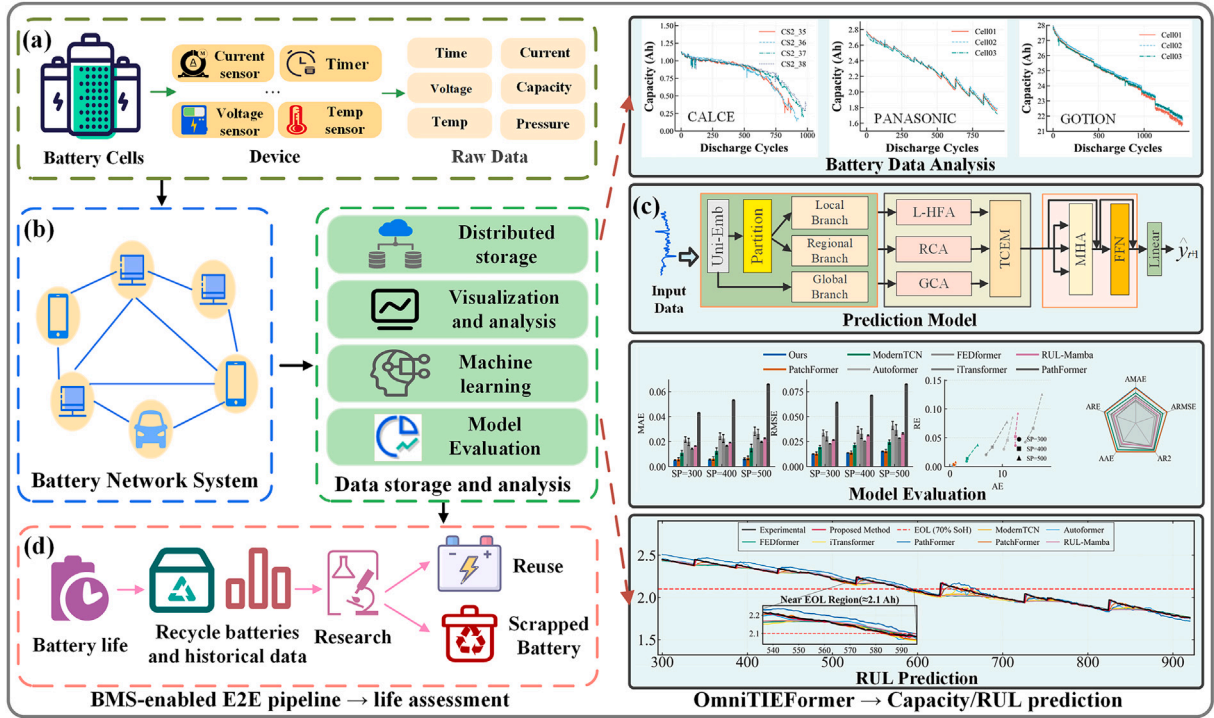


Fig. 1. The proposed End-to-End (E2E) framework for battery lifecycle management, illustrating four key stages: (a) data acquisition, (b) cloud-based analysis, (c) capacity trajectory forecasting and RUL estimation via the OmniTIEFormer model, and (d) BMS-enabled decisions that determine the battery's subsequent value chain, including second-life applications, research, or material recycling.

increasingly intricate pipelines, exemplified by Fei et al. [22], which constructed a framework with 42 handcrafted features. More broadly, feature-engineered pipelines can be constrained by the completeness and cross-dataset generalizability of the chosen descriptors, which may limit gains when key degradation factors are not adequately captured. These considerations have motivated growing interest in end-to-end models that learn task-relevant representations directly from raw or minimally processed measurements [23].

Advanced deep learning architectures have propelled RUL prognostics across industrial and electromechanical systems [24], yet direct adoption in batteries remains challenging because degradation is highly nonlinear and governed by coupled dynamics across time scales. This has driven a progression from RNN-based models [25,26] to Transformer backbones [27,28]. In parallel, temporal deep learning has begun to incorporate uncertainty estimation to improve the reliability of health-state inference [29]. Building on these backbones, hybrid designs have emerged, such as the BiGRU-Transformer model of Jia et al. [30], and architectural modifications, including Dilated Causal Convolution (DCC), to strengthen local feature extraction [31]. More recently, parallel Transformer structures fused with Kolmogorov–Arnold Networks (KAN) have been explored to further enhance state-of-health prediction [32]. Despite these advances, many end-to-end models still treat multi-scale information (e.g., local cycle-to-cycle patterns versus long-horizon degradation trends) in largely separate ways, either through serial feed-forward pipelines or parallel branches with limited interaction. This design choice can make it difficult to capture concurrent and entangled degradation behaviors.

This challenge of modeling multi-scale dynamics has been addressed but remains incompletely resolved. Some methods transform raw data into intermediate representations, such as the graphical features used by He et al. [33] for CNNs, but this can still enforce a largely serial extract-then-predict workflow. Similar limitations may persist in more advanced architectures: the multi-scale CNN of Yang et al. [34]

aggregates scale-specific features primarily at later stages, thereby limiting real-time cross-scale feedback. Similarly, the TOP-Net proposed by Chen et al. [14] adopts a parallel 2D–1D hybrid architecture to extract multi-scale degradation features under diverse cycling protocols, demonstrating improved cross-profile generalization. Nevertheless, branch interaction is limited to output-stage concatenation, with no information exchange during feature learning; consequently, each branch independently optimizes its own representation, without benefiting from complementary scale information. Meanwhile, the multi-resolution model of Geng et al. [35] adopts a largely unidirectional flow in which high-level context cannot readily refine low-level interpretations. We address this limitation by shifting from serial processing to concurrent, interactive modeling [36]. Specifically, we propose OmniTIEFormer, whose core component, the TCEM, enables real-time bidirectional information exchange between parallel streams, promoting synergy between local phenomena and global trends.

Data scarcity in practical settings has further spurred the development of data-efficient learning strategies. Semi-supervised learning has been explored to leverage unlabeled data for few-shot prediction [37], although such methods can introduce additional training complexity and may exhibit variable transfer behavior across heterogeneous datasets. Transfer learning is another widely adopted approach [38,39]; for example, the Pre-trained Predictive Transformer (PPT) proposed by Zhao et al. [13] achieves strong performance, yet its predominantly unidirectional information flow can be limiting in some transfer settings. When transferred to cells with distinct recovery dynamics, fixed pre-trained macro-trends may not be adaptively refined by emerging local patterns, increasing the risk of *negative transfer*. These observations point to a shared bottleneck: both semi-supervised and transfer learning benefit from backbones that can encode robust, transferable multi-scale representations. OmniTIEFormer is designed toward this objective, with the aim of better unlocking the potential of data-efficient strategies.

Table 1

Structured comparison of representative architectures for battery degradation modeling, highlighting scale modeling capabilities and cross-scale interaction mechanisms.^a

Method	Architecture	Scale modeling	Cross-scale interaction	Remaining limitation
<i>Group I: Sequential / single-scale</i>				
LSTM/GRU [25,26]	Sequential RNN	Single-scale	Not explicitly modeled	Temporal patterns captured within a single latent representation, without explicit scale separation.
BiGRU-Trans. [40]	RNN → Transformer	Dual-scale (serial)	One-way cascade	Global Transformer representation mainly refines RNN features, with limited feedback to earlier layers.
<i>Group II: Multi-scale with limited interaction</i>				
Graphical CNN [33]	CNN on graphical features	Multi-view (CV/IC/CD)	Sequential pipeline	Multi-view degradation cues are represented as images, with limited joint modeling in sequence space.
PPT [13]	CNN + sparse Transformer	Multi-scale (serial)	Unidirectional aggregation	Feature extraction and aggregation are performed sequentially, with limited revision from global trends to local signals.
Multi-res. Agg. [35]	Hierarchical network	Multi-resolution	Top-down only	Hierarchical context aggregation primarily flows from higher to lower levels.
<i>Group III: Parallel / multi-branch with post-hoc fusion</i>				
TOP-Net [14]	2D + 1D parallel NN	Multi-scale (parallel)	Output-stage concatenation	Parallel branches interact primarily at the output stage.
Parallel Trans. + KAN [32]	Parallel Trans. + KAN	Multi-scale (parallel)	KAN-based fusion	Cross-branch coordination mainly occurs during the fusion stage.
PatchFormer [41]	Patch-based Transformer	Intra-/inter-patch	Patch-level interaction	Patch hierarchy captures local-to-global context without an explicit regional-scale branch.
PathFormer [42]	Adaptive multi-scale Trans.	Multi-scale (adaptive)	Routing-based	Routing dynamically selects among scales but does not explicitly exchange representations across branches.
OmniTIEFormer (ours)	Tri-branch Transformer	Concurrent L-R-G	TCEM-based bidirectional exchange	The current design uses three predefined scales; adaptive scale selection remains a potential extension.

^a L-R-G = local-regional-global. TCEM = Tri-branch Cross-Exchange Module. Trans. = Transformer. CV = constant-voltage; IC = incremental capacity; CD = charge-discharge.

In summary, existing battery prognostics models, whether serial or parallel, often rely on unidirectional or weakly coupled information flows, limiting the dynamic interactions needed to capture complex degradation physics. This limitation can also impair representation robustness, thereby reducing the effectiveness of data-efficient learning, including transfer learning. To systematically delineate this architectural gap, Table 1 compares representative methods along four design dimensions: backbone architecture, scale modeling strategy, cross-scale interaction mechanism, and the remaining limitation each method leaves unaddressed. The methods are organized into three groups that trace the evolution of architectural paradigms: (I) sequential single-scale models that compress all temporal information into a single representation, (II) multi-scale architectures that decompose degradation dynamics but restrict cross-scale interaction to unidirectional or post-hoc aggregation, and (III) parallel multi-branch designs that process scales concurrently yet still defer branch communication to a single fusion point after independent feature extraction. As Table 1 reveals, a persistent bottleneck spans all three groups: cross-scale information exchange, when present, is universally restricted to a single post-hoc stage. Serial designs such as PPT [13] aggregate scale-specific features in a feed-forward manner, while parallel architectures including TOP-Net [14] and Parallel Trans.+KAN [32] fuse branch outputs only after independent processing. None of these architectures enables a global-trend representation to dynamically correct a local branch's misinterpretation of noise as regeneration, or vice versa, *during* the feature learning process itself. A more general solution, therefore, calls for architectures that simultaneously support: (1) concurrent multi-scale processing through dedicated branches, (2) explicit bidirectional cross-scale information exchange during representation learning, and (3) transferable representation learning across battery chemistries and capacity scales.

3. Methodology

This section first formulates the battery prognostics task and clarifies the use of capacity as the sole model input. We then present OmniTIEFormer, an **Omni-scale Tri-branch Information Exchange Transformer** designed to capture battery degradation dynamics across multiple temporal scales. The model employs three parallel branches to encode local, regional, and global temporal dependencies, while a TCEM enables dynamic information interaction across scales. The fused representation is subsequently processed by a lightweight Transformer-based temporal encoder for single-step capacity prediction.

3.1. Problem formulation and capacity-only input

We formulate battery prognostics in this study as a sliding-window, one-step-ahead capacity-forecasting problem over historical capacity sequences. Given a fixed-length input window,

$$\mathbf{X}_{\text{history}} = \{c_{t-L+1}, c_{t-L+2}, \dots, c_t\},$$

the model is trained to predict the capacity of the next cycle:

$$f(\mathbf{X}_{\text{history}}) \rightarrow \hat{c}_{t+1}. \quad (1)$$

The supervised objective is limited to next-cycle capacity prediction. During inference, the one-step predictor is recursively rolled out by feeding each predicted output back into the input window to generate the future degradation trajectory. The predicted EOL cycle, denoted by $\hat{N}_{\text{EOL}}$, is estimated by linearly interpolating the first threshold-crossing segment of the forecasted capacity trajectory. The RUL at cycle t is then estimated as

$$\widehat{RUL}_t = \hat{N}_{\text{EOL}} - t. \quad (2)$$

Capacity is selected as the sole input feature for two practical reasons. First, compared with raw voltage or current signals, capacity more directly reflects the long-term degradation trajectory and is generally less sensitive to short-term operational fluctuations and measurement artifacts. Second, using a single scalar sequence reduces input dimensionality, decreases model complexity and computational cost, and mitigates overfitting risks associated with high-dimensional handcrafted feature spaces. Consequently, the proposed OmniTIEFormer learns directly from historical capacity curves, avoiding extensive manual feature engineering while establishing a streamlined, transferable prognostic framework.

3.2. Model architecture overview

As illustrated in Fig. 2, OmniTIEFormer consists of four main stages. Before detailing them, we first clarify the design rationale and physical motivation of the tri-branch architecture.

Design rationale and physical motivation. Battery degradation is governed by coupled processes operating over multiple temporal scales rather than a single homogeneous timescale. At the short-term scale, capacity trajectories may exhibit local regeneration phenomena, transient disturbances, and measurement noise caused by rest periods or operational fluctuations. At the intermediate scale, batteries often experience stage-wise evolution, such as changes in fading rate, staircase-like degradation segments, or localized non-monotonic behavior. At the long-term scale, irreversible aging mechanisms, including solid electrolyte interphase (SEI) growth, loss of lithium inventory (LLI), and loss of active material (LAM), dominate the overall degradation trend.

The tri-branch architecture is therefore designed to explicitly encode these three complementary temporal scopes. Specifically, the local branch captures short-term fluctuations within small temporal neighborhoods, the regional branch emphasizes intermediate structural evolution in degradation trajectories, and the global branch models the overall long-horizon degradation trend. Rather than assuming a

strict one-to-one mapping between each branch and a single electrochemical mechanism, the branches are intended to capture degradation patterns primarily shaped by physical processes acting over similar temporal scales. In this sense, the tri-branch design serves as a practically effective decomposition of battery degradation dynamics: a single-branch architecture may entangle heterogeneous temporal patterns, whereas a two-branch design may still merge intermediate transitions with either short-term perturbations or long-term drift.

(i) **Embedding.** A univariate input sequence is projected into a C -dimensional latent space, yielding $\mathbf{X}_{\text{emb}} \in \mathbb{R}^{B \times L \times C}$.

(ii) **Parallel multi-scale feature recalibration.** We deploy three parallel pathways to capture heterogeneous temporal dynamics across multiple contextual scopes: fine-grained local structures, mid-range contextual interactions, and global context anchoring via channel-wise recalibration. The local and regional branches first operate on patch-level representations, while the global branch remains at the original temporal resolution. Before entering TCEM, the branch outputs are aligned in the same feature space, $\mathbb{R}^{B \times C \times L}$, to enable seamless cross-scale fusion.

(iii) **Tri-branch cross-exchange module (TCEM).** TCEM enables cross-scale interaction among the three branches. Specifically, the local and regional branches exchange information via channel-wise cross-attention, while the global branch integrates the contextual cues generated during cross-scale interaction. The enhanced representations from all branches are then concatenated along the temporal dimension, followed by channel recalibration and temporal folding to restore the original sequence length.

(iv) **Temporal encoding and prediction.** A lightweight Transformer-based temporal encoder processes the TCEM output, which is subsequently mapped by a linear prediction head to produce the single-step capacity prediction $\hat{c}_{t+1}$.

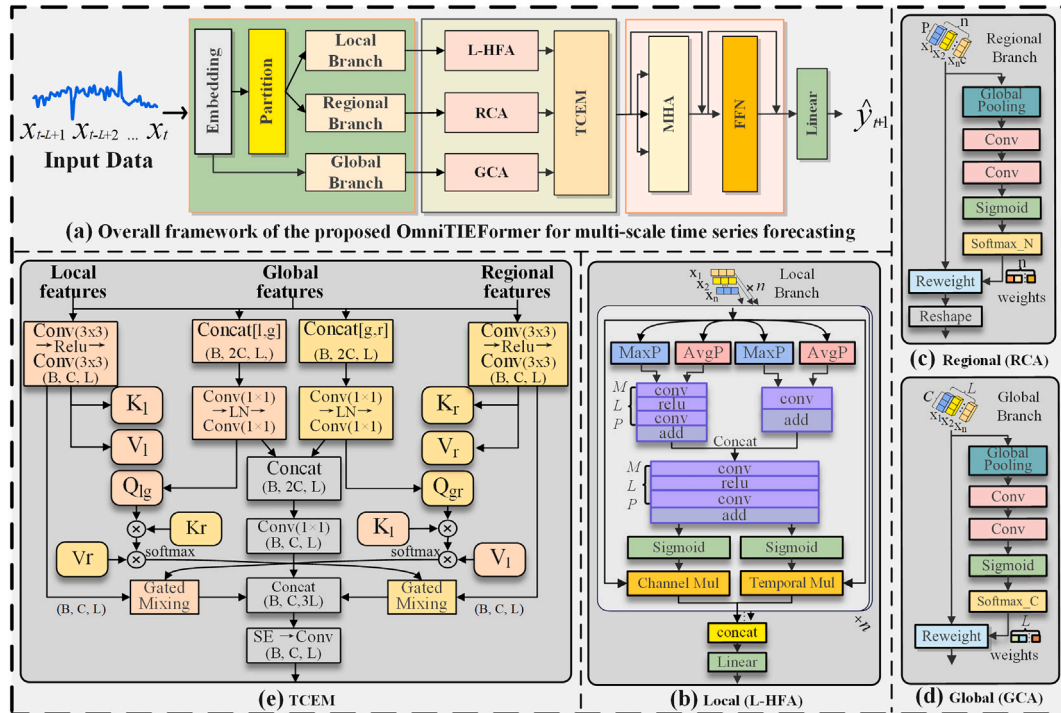


Fig. 2. The architecture of OmniTIEFormer. (a) Overall framework for multi-scale time-series forecasting; (b) Local Hybrid Fusion Attention (L-HFA); (c) Regional Channel Attention (RCA); (d) Global Channel Attention (GCA); (e) Tri-branch Cross-Exchange Module (TCEM).

3.3. Input representation and embedding

We consider univariate forecasting with input

$$\mathbf{x} \in \mathbb{R}^{B \times L}.$$

A linear projection maps each scalar observation into a C -dimensional embedding space:

$$\mathbf{X}_{\text{emb}} = \text{Embed}(\mathbf{x}) \in \mathbb{R}^{B \times L \times C}. \quad (3)$$

For downstream modules, we adopt a channel-first representation:

$$\mathbf{X}_{\text{emb}}^T \in \mathbb{R}^{B \times C \times L}.$$

Patch notation. For local and regional operations, we define a patch length P and the number of non-overlapping patches $N = L/P$. If P does not divide L , the sequence is padded to the nearest multiple of P , and the padding is removed after processing. The *Unfold* operation segments a sequence into patches of length P , while the *Fold* operation restores the original temporal structure.

3.4. Parallel multi-scale feature recalibration

Conventional sequential or hierarchical modeling approaches often impose rigid abstraction stages, making it difficult to preserve both fine-grained details and global semantics simultaneously. To address this limitation, we introduce three parallel residual branches that explicitly model temporal dependencies at different contextual scales: local dynamics, regional interactions, and global sequence-wide characteristics.

The three branches share a common latent representation space while focusing on different temporal scopes. The local and regional branches first operate on patch-level representations, whereas the global branch remains at the original temporal resolution. This design preserves architectural homogeneity while aligning branch outputs before cross-scale fusion in TCEM. This design is motivated by the observation that time-series dynamics typically arise from a combination of short-term fluctuations (local), event-driven or seasonal trends (regional), and global stationary properties. Attempting to capture these heterogeneous patterns with a single-path hierarchical architecture may lead to mutual interference and information loss. The proposed parallel design, therefore, allows each branch to specialize in its corresponding temporal scope.

3.4.1. Local branch: local hybrid fusion attention (L-HFA)

To model fine-grained dynamics within each temporal segment, we propose L-HFA. Unlike cascaded attention blocks (e.g., CBAM) that calibrate different domains sequentially, L-HFA performs parallel descriptor extraction and joint cross-domain coupling through a lightweight fusion bottleneck, as illustrated in Fig. 2(b).

Anchor-augmented joint modulation. For each patch of length P , we prepend a learnable anchor to the patch tokens, forming an augmented segment $\tilde{\mathbf{U}} \in \mathbb{R}^{B \times C \times (P+1)}$. Rather than computing and applying temporal and channel gates independently, L-HFA extracts initial descriptors via parallel pooling and fuses them using a shared multi-layer perceptron ($\text{MLP}_{\text{fusion}}$). Specifically,

$$\begin{aligned} [\tilde{\mathbf{w}}^{(c)}, \tilde{\mathbf{w}}^{(t)}] &= \text{Split}(\text{MLP}_{\text{fusion}}([\text{Pool}_C(\tilde{\mathbf{U}}); \text{Pool}_{P+1}(\tilde{\mathbf{U}})])), \\ \tilde{\mathbf{U}} &= \tilde{\mathbf{U}} \odot \tilde{\mathbf{w}}^{(c)} \odot \tilde{\mathbf{w}}^{(t)} + \tilde{\mathbf{U}}. \end{aligned} \quad (4)$$

The updated anchor token (the first position of $\tilde{\mathbf{U}}$) is extracted as the compact local representation for each patch. Collecting these anchor tokens across all patches yields the local branch representation, which is subsequently aligned to the original temporal resolution before entering TCEM.

How L-HFA differs from standard attention/SE:

- **Anchor readout:** It summarizes each segment via a learnable representative token, rather than only reweighting existing features.
- **Cross-domain coupling:** Temporal and channel cues are extracted in parallel and *jointly refined* through the fusion MLP, avoiding the dependency bottleneck of sequential CBAM-style gating.
- **Lightweight modulation:** It enables local adaptation without the computational overhead of full pairwise token interactions.

3.4.2. Regional branch: regional channel attention (RCA)

The regional branch highlights which temporal segments (patches) are most informative for forecasting. Although termed Regional Channel Attention (RCA) for consistency with the overall channel-first implementation, this module actually performs **patch-wise recalibration** over the segment axis rather than conventional feature-channel recalibration, and it enforces **competition** across segments, as depicted in Fig. 2(c).

Let $\mathbf{V} \in \mathbb{R}^{B \times C \times N}$ denote the aggregated patch sequence, where $N = L/P$ is the number of patches. RCA first squeezes the channel axis to obtain a patch-level descriptor $\mathbf{s} = \text{AvgPool}_C(\mathbf{V}) \in \mathbb{R}^{B \times 1 \times N}$. It then applies an SE-style bottleneck followed by an explicit **Softmax across patches** to produce competitive saliency scores:

$$\mathbf{w}^{(\text{patch})} = \text{Softmax}_N(\psi(\mathbf{s})) \in \mathbb{R}^{B \times 1 \times N}, \quad (5)$$

where $\psi(\cdot)$ is a lightweight squeeze-excitation MLP. Finally, $\mathbf{V}$ is reweighted patch-wise by $\mathbf{w}^{(\text{patch})}$. The resulting regional representation is then aligned to the original temporal resolution before being passed to TCEM.

How RCA differs from standard SE:

- **Dimensional shift (When vs. What):** Standard SE recalibrates *feature channels* (C), whereas RCA recalibrates *patch positions* (N), acting as a regional saliency filter over time.
- **Competitive selection:** Softmax across N forces the model to prioritize the most discriminative segments, in contrast to a Sigmoid gate that independently weights each feature.

3.4.3. Global branch: global channel attention (GCA)

While the local and regional branches focus on fine-grained transient dynamics and mid-range trend saliency, the global branch embeds sequence-wide stationary properties, such as the battery's overall degradation stage, into the feature representation, as shown in Fig. 2(d).

To achieve this holistic view without adding unnecessary complexity, GCA extracts a global context vector by compressing the temporal dimension L , broadcasts it back to the original sequence length, and applies channel-wise Squeeze-and-Excitation (SE_C) recalibration:

$$\mathbf{X}_g = \text{SE}_C\left(\text{Expand}_L\left(\text{AvgPool}_L(\mathbf{X}_{\text{emb}}^T)\right)\right) + \mathbf{X}_{\text{emb}}^T. \quad (6)$$

How GCA differs from standard SE:

- **Role in the multi-scale framework:** Unlike standard SE usage which produces dynamic weights from local receptive fields, GCA acts as a *stationary global anchor*.
- **Macroscopic context:** By broadcasting the temporally pooled summary $\text{AvgPool}_L(\cdot)$ across all time steps, GCA keeps each local representation aware of the global degradation trajectory, reducing the risk of overfitting to localized fluctuations.

Hierarchical multi-scale fusion. The three feature streams capture complementary temporal characteristics: $\mathbf{X}_l$ focuses on fine-grained local patterns, $\mathbf{X}_r$ emphasizes informative regional segments, and $\mathbf{X}_g$ encodes global contextual awareness. After patch-level processing in the local and regional branches, their outputs are folded back to the original temporal resolution so that all three branches are represented in the same space, $\mathbb{R}^{B \times C \times L}$, before entering TCEM. For the local branch,

the anchor representation extracted from each patch is broadcast to the corresponding P temporal positions; for the regional branch, the patch-wise saliency-weighted representation is expanded in the same manner. This produces $\mathbf{X}_l, \mathbf{X}_r \in \mathbb{R}^{B \times C \times L}$ before TCEM. This alignment enables direct cross-scale interaction while preserving local sensitivity, regional selectivity, and global trend information within a unified temporal representation.

3.5. Tri-branch cross-exchange module (TCEM)

To enable dynamic, context-aware fusion of multi-scale representations, we design the TCEM. This module facilitates gated information exchange between the local and regional branches while allowing the global branch to absorb cross-scale interaction cues. An overview of this architecture is illustrated in Fig. 2(e).

Step 1: Self-enhancement. The local and regional branches first undergo a specialized 1D convolutional enhancement block (configured with a kernel size of 3) to extract deeper scale-specific features. Meanwhile, the global features remain intact for subsequent broadcasting:

$$\hat{\mathbf{X}}_s = f_{\text{self}}(\mathbf{X}_s), \quad s \in \{l, r\}. \quad (7)$$

Step 2: Query generation. The global features are concatenated with the enhanced local and regional features. According to this design, these joint representations are processed by a projection block ($\text{Conv}_{1 \times 1} \rightarrow \text{LN} \rightarrow \text{Conv}_{1 \times 1}$) to generate the cross-branch queries:

$$\mathbf{Q}_l = f_q([\hat{\mathbf{X}}_l; \mathbf{X}_g]), \quad \mathbf{Q}_{gr} = f_q([\mathbf{X}_g; \hat{\mathbf{X}}_r]). \quad (8)$$

Step 3: Key-value mapping. Standard linear mappings extract keys and values from the enhanced local and regional streams:

$$\mathbf{K}_l = f_k(\hat{\mathbf{X}}_l), \quad \mathbf{V}_l = f_v(\hat{\mathbf{X}}_l), \quad \mathbf{K}_r = f_k(\hat{\mathbf{X}}_r), \quad \mathbf{V}_r = f_v(\hat{\mathbf{X}}_r). \quad (9)$$

Step 4: Channel-wise cross-attention. Using the generated queries, the local branch attends to the regional branch's features, and vice versa. In implementation, each feature map in $\mathbb{R}^{B \times C \times L}$ is reshaped into a batch of $C \times L$ matrices, where each channel is treated as a length- L vector. Channel-to-channel similarities are then computed between the query and key matrices, and the Softmax is applied over the key-channel dimension. The resulting cross-attention therefore acts along the channel dimension:

$$\mathbf{A}_{l \leftarrow r} = \text{Softmax}_C(\tau \mathbf{Q}_l \mathbf{K}_r^T) \in \mathbb{R}^{B \times C \times C}, \quad \mathbf{Z}_{l \leftarrow r} = \mathbf{A}_{l \leftarrow r} \mathbf{V}_r \in \mathbb{R}^{B \times C \times L}, \quad (10)$$

where τ is a learnable temperature coefficient that adaptively scales the attention logits. The symmetric operation yields $\mathbf{Z}_{r \leftarrow l}$ using $\mathbf{Q}_{gr}$, $\mathbf{K}_l$, and $\mathbf{V}_l$.

Step 5: Gated mixing. The cross-attention outputs are fused with their respective original enhanced features via a learnable gating mechanism:

$$\tilde{\mathbf{X}}_l = g_{l \leftarrow r} \odot \mathbf{Z}_{l \leftarrow r} + (1 - g_{l \leftarrow r}) \odot \hat{\mathbf{X}}_l, \quad g_{l \leftarrow r} = \sigma(W(\mathbf{Z}_{l \leftarrow r} + \hat{\mathbf{X}}_l)), \quad (11)$$

and analogously for $\tilde{\mathbf{X}}_r$.

Step 6: Global update. To synchronize the global context with the newly exchanged information, the intermediate query representations are concatenated along the channel dimension (yielding $B \times 2C \times L$) and projected back to the original channel size via a 1×1 convolution. A residual connection is employed to preserve the intact global baseline:

$$\hat{\mathbf{X}}_g = \text{Conv1d}_{1 \times 1}([\mathbf{Q}_l; \mathbf{Q}_{gr}]) + \mathbf{X}_g. \quad (12)$$

Step 7: In-module fusion and temporal folding. The three thoroughly updated streams ($\tilde{\mathbf{X}}_l, \tilde{\mathbf{X}}_g, \tilde{\mathbf{X}}_r$) are concatenated along the temporal axis to form an extended sequence:

$$\mathbf{X}_{\text{cat}} = [\tilde{\mathbf{X}}_l; \tilde{\mathbf{X}}_g; \tilde{\mathbf{X}}_r] \in \mathbb{R}^{B \times C \times 3L}. \quad (13)$$

This concatenated sequence undergoes a final channel-wise Squeeze-and-Excitation (SE_C) recalibration. Finally, to project the sequence back to the original length L while fusing the multi-scale temporal information, we apply a temporal folding 1D convolution (configured with a kernel size and stride of 3):

$$\mathbf{X}_{\text{tce}} = \text{Conv1d}_{k=3, s=3}(\text{SE}_C(\mathbf{X}_{\text{cat}})) \in \mathbb{R}^{B \times C \times L}. \quad (14)$$

3.6. Temporal encoding and single-step prediction

Following multi-scale feature fusion within TCEM, the resulting representation $\mathbf{X}_{\text{tce}}$ provides a context-aware description at each time step. The subsequent lightweight Transformer encoder is used not to re-extract multi-scale features, but to model the long-range temporal dependencies among these already-enhanced representations. Let $\mathbf{H}^{(0)} = \mathbf{X}_{\text{tce}}$. A lightweight Transformer with E layers encodes temporal dependencies over L steps:

$$\mathbf{H}' = \mathbf{H} + \text{MHA}(\mathbf{H}), \quad (15)$$

$$\mathbf{H}^+ = \text{LN}(\mathbf{H}' + f_{\text{FFN}}(\mathbf{H}')), \quad (16)$$

where MHA, LN, and f_{FFN} denote multi-head attention, layer normalization, and the feed-forward network, respectively. This procedure is applied layer-wise. Unless otherwise stated, attention is full (non-causal); a triangular causal mask is used only in strictly autoregressive settings.

Prediction head (single-step). Flatten (L, C) and apply a linear projection:

$$\hat{c}_{t+1} = \mathbf{w}^T \text{vec}(\mathbf{H}^+) + b, \quad \hat{c}_{t+1} \in \mathbb{R}^{B \times 1}. \quad (17)$$

This head has $\mathcal{O}(LC)$ parameters and computes the final single-step prediction.

3.7. Training objective: mean absolute error (MAE)

Given the ground-truth capacities $\{c_{t+1}^{(i)}\}_{i=1}^B$ and the predictions $\{\hat{c}_{t+1}^{(i)}\}_{i=1}^B$, we adopt the mean absolute error as the training objective:

$$\mathcal{L}_{\text{MAE}} = \frac{1}{B} \sum_{i=1}^B |c_{t+1}^{(i)} - \hat{c}_{t+1}^{(i)}|. \quad (18)$$

MAE is more robust to outliers, provides stable gradients in noisy time series, and aligns with standard evaluation metrics.

3.8. Methodological summary and end-to-end formulation

OmniTIEFormer consists of four core components. First, a univariate capacity sequence is mapped into a unified latent representation space through an embedding layer. Second, three parallel branches, namely local, regional, and global branches, are used to capture fine-grained fluctuations, mid-range evolutionary dynamics, and long-term degradation trends, respectively. Third, the TCEM module enables cross-scale interaction, dynamic fusion, and global updating among these multi-scale representations. Finally, a lightweight Transformer-based temporal encoder and a linear prediction head are used to produce the capacity prediction for the next cycle. Based on this single-step prediction, the model further generates the future capacity evolution trajectory through recursive inference, from which the EOL point and the RUL are estimated.

From a methodological standpoint, OmniTIEFormer can be formulated as an end-to-end predictive model that maps historical capacity sequences directly to next-cycle capacity prediction and, through recursive inference, to lifetime estimation results. The model takes the historical capacity sequence as input and, under a unified optimization objective, jointly performs representation learning and cross-scale fusion for next-cycle capacity forecasting, from which lifetime indicators are estimated via recursive inference. In this way, it avoids reliance on manually designed health-related features. It departs from the fragmented multi-stage workflow commonly found in conventional approaches, where feature extraction, state estimation, and lifetime regression are treated as separate procedures. This formulation enables the model to learn task-relevant latent representations adaptively from the capacity evolution process, thereby reducing dependence on prior feature engineering. At the same time, it mitigates the information fragmentation and error accumulation that may arise in multi-stage pipelines, allowing local fluctuations, regional transitions, and global degradation trends to be modeled collaboratively under a single objective. Because the model does not depend on a predefined feature set, it is also less sensitive to feature completeness and the quality of manually constructed indicators. This property gives the model greater flexibility under different data conditions and provides a more direct representational foundation for cross-dataset transfer learning.

4. Evaluation

4.1. Experimental datasets

To comprehensively evaluate the performance, generalization capability, and robustness of the proposed prognostics model, three publicly available and heterogeneous lithium-ion battery cycling aging datasets were selected: CALCE [43], PANASONIC [15], and GOTION HIGHTECH [15]. These datasets span markedly different chemistries, capacities, operating temperatures, and charge/discharge protocols, covering use cases from small portable devices to large-scale traction and energy-storage systems. Key parameters are summarized in Table 2, followed by dataset-specific details. The EOL thresholds used for RUL labeling follow common conventions in battery health prognosis literature (typically 70% or 80% of the nominal capacity). They are specified according to the characteristics of each dataset.

4.1.1. CALCE dataset

The CALCE dataset was released by the Center for Advanced Life Cycle Engineering at the University of Maryland. It contains prismatic CS2-series cells with an LCO cathode (LiCoO_2) and graphite anode, rated at 1.1 Ah. Four cells (CS2_35, CS2_36, CS2_37, and CS2_38) were used. All tests were conducted at an extremely low temperature of 1 °C. Charging followed a CC–CV protocol: a 0.5C constant current (0.55 A) was applied until the voltage reached 4.2 V, followed by constant-voltage charging until the current decreased to a minimal threshold. Discharging was performed at 1C (1.1 A) constant current with a cutoff voltage of 2.7 V. For each cycle, the measured discharge capacity was

extracted to construct the capacity degradation sequence used for model training and evaluation.

4.1.2. PANASONIC dataset

The PANASONIC dataset involves cylindrical NCR18650BD cells with an NCA cathode (LiNiCoAlO_2) and a graphite anode, rated at 3.03 Ah. Three cells were analyzed. Tests were carried out at 20 °C using an ARBIN BT2000 system. Charging followed a CC–CV protocol: a 0.3C constant current (0.91 A) was applied until the voltage reached 4.2 V, followed by constant-voltage charging until the current fell below 0.03 A. Discharging was performed at 2C (6.06 A) constant current with a cutoff voltage of 2.5 V. As with CALCE, the measured capacity value of each cycle was extracted to construct the degradation trajectory used in this study.

4.1.3. GOTION dataset

The GOTION dataset consists of large prismatic IFP20100140A cells with an LFP cathode (LiFePO_4) and a graphite anode, rated at 27 Ah. Three cells were selected. Tests were conducted at 45 °C using an ARBIN BT2000 system. Charging followed a CC–CV protocol: a 1C constant current (27 A) was applied until the voltage reached 3.65 V, followed by constant-voltage charging until the current decreased below 1.5 A (0.05C). Discharging was performed at 1C (27 A) constant current with a cutoff voltage of 2.0 V. For each cycle, the measured capacity value was extracted to construct the capacity degradation sequence for analysis.

In battery RUL research, the EOL threshold is commonly defined as a percentage of the nominal capacity, with 70% or 80% being the most widely adopted criteria in the literature. For small-capacity cells such as those in the CALCE and PANASONIC datasets, the EOL threshold is set to 70% of the nominal capacity. However, the large-capacity 27 Ah LFP cells in the GOTION dataset degrade more slowly and do not reach 70% state of health within the available cycling data. Therefore, following common practice for large-format batteries, the EOL threshold is set at 80% of the nominal capacity, corresponding to 21.6 Ah for RUL labeling.

4.2. Data preparation

4.2.1. Data split, sample construction, and normalization

Data split and sample construction. We evaluate generalization using a cell-disjoint hold-out protocol. For each dataset, one cell is reserved in its entirety for testing, and the remaining cells are used for model development (Table 3). To prevent temporal leakage, each training/validation cell is split chronologically: the first 80% of its degradation trajectory is used for training, and the final 20% is reserved exclusively for validation and early stopping.

Time-series inputs are generated by sliding-window sampling over the cycling trajectories with a window length of 64 cycles, a prediction horizon of 1 cycle, and a stride of 1. The window length of 64 was deliberately selected for two primary reasons. Physically, 64 cycles represent approximately one-tenth of the total lifecycle for a standard CALCE cell, providing a moderate historical snapshot that captures meaningful degradation trends without introducing excessive sequence redundancy. Methodologically, it aligns with the mainstream standard adopted by

Table 2
Key parameters of the experimental datasets.

Parameter	CALCE [43]	PANASONIC [15]	GOTION [15]
Cell model	CS2 Series	NCR18650BD	IFP20100140A
Cathode/Anode	LiCoO_2 /Graphite (LCO)	NCA/Graphite	LiFePO_4 /Graphite (LFP)
Format	Prismatic	Cylindrical	Prismatic
Rated capacity	1.1 Ah	3.03 Ah	27 Ah
Voltage range	2.7 V to 4.2 V	2.5 V to 4.2 V	2.0 V to 3.65 V
Test temperature	1 °C (low temperature)	20 °C (room temperature)	45 °C (high temperature)
Charging strategy	0.5C CC–CV to 4.2 V	0.3C CC–CV to 4.2 V	1C CC–CV to 3.65 V
Discharging strategy	1C CC, cutoff 2.7 V	2C CC, cutoff 2.5 V	1C CC, cutoff 2.0 V
Number of cells used	4 cells	3 cells	3 cells

Table 3

Dataset division and generated sample sizes.

Dataset	Train & Val Cells	Test Cell	EOL	Samples (Tr/Va/Te)	Total
CALCE	CS2_36, 37, 38	CS2_35	70%	2131 / 517 / 583	3231
PANASONIC	Cell 02, 03	Cell 01	70%	1352 / 306 / 471	2129
GOTION	Cell 02, 03	Cell 01	80%	2145 / 505 / 972	3622

Note: Tr, Va, and Te denote the number of samples for training, validation, and testing, respectively.

recent state-of-the-art baselines (e.g., PatchFormer, RUL-Mamba), ensuring a strictly fair benchmark comparison. This procedure effectively increases the number of training instances despite the limited number of physical cells, while strictly preserving the causal ordering of measurements. Furthermore, to ensure the model learns from genuine degradation dynamics, no additional artificial data augmentation or resampling strategy was applied. The total number of cycles per cell and the resulting counts of generated samples for training, validation, and testing are reported in Table 3.

Inter-cell variability and normalization. To mitigate inter-cell variability arising from manufacturing tolerances, we apply min-max normalization to the battery capacity sequences. Specifically, the capacity values are scaled into the range [0, 1] according to

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}. \quad (19)$$

Since the datasets used in this study have substantially different nominal capacities, normalization is performed independently for each dataset. The scaling parameters ($x_{\min}$ and $x_{\max}$) are computed using only the training cells under the cell-disjoint hold-out protocol and are then applied to the corresponding validation and test cells. This dataset-wise normalization prevents information leakage while ensuring consistent scaling across cells within the same dataset.

Furthermore, sliding windows are generated independently for each physical cell, and no sample crosses cell boundaries. The model takes only the capacity sequence as input and does not use any cell identifier. Together with the cell-disjoint hold-out split and normalization parameters estimated only from the training cells, this design prevents information leakage between the training, validation, and test sets. No additional data augmentation, resampling, or class-rebalancing strategy was applied beyond the sliding-window sampling procedure described above.

4.2.2. Evaluation protocol

RUL evaluation in this study follows the task formulation introduced in Section 3.1. Specifically, the model first performs single-step capacity forecasting, and the future degradation trajectory is then generated recursively until the predicted capacity reaches the dataset-specific EOL threshold. The predicted EOL cycle is used to compute the corresponding RUL.

To assess performance at different degradation stages, we evaluate recursive forecasting from multiple starting points. For CALCE and PANASONIC, the starting points are set to 300, 400, and 500 cycles. For GOTION, the starting points are set to 450, 600, and 750 cycles. These settings allow evaluation under early-, middle-, and late-stage degradation conditions.

4.3. Benchmark methods

To rigorously evaluate the effectiveness of OmniTIEFormer, we compare it with several strong and widely recognized baselines from both the general time-series forecasting (TSF) community and recent battery RUL prediction studies. The selected models represent diverse architectural paradigms, including convolutional networks, Transformer-based

methods, and state-space models. This diversity ensures a comprehensive evaluation against competitive approaches with different temporal modeling mechanisms.

General-purpose TSF models. We include several high-performing models from the broader TSF literature. **ModernTCN** [44] represents convolutional temporal modeling and serves as a strong CNN-based baseline for extracting local temporal patterns. **Autoformer** [45] and **FEDformer** [46] are representative Transformer variants designed for long-sequence forecasting. **iTransformer** [47] introduces an inverted attention mechanism and serves as a strong Transformer-based baseline for temporal dependency modeling, while **PathFormer** [42] adopts a multi-scale Transformer architecture that provides an additional point of comparison for hierarchical temporal modeling.

Battery RUL prediction models. To benchmark against recent domain-specific advances, we also include two state-of-the-art models specifically designed for battery prognostics. **PatchFormer** [41] is a Transformer-based architecture tailored for battery capacity degradation modeling and has demonstrated strong performance in recent studies. **RUL-Mamba** [48] represents a recent trend toward state-space sequence modeling in battery RUL prediction, offering an alternative architecture capable of capturing long-range degradation dynamics. Including these specialized models enables direct comparison with the latest approaches to battery health prediction.

4.4. Evaluation metrics

We evaluate curve-fitting accuracy (MAE, RMSE, R^2) and RUL prediction accuracy (AE, RE). The predicted EOL cycle, $\hat{N}_{\text{EOL}}$, is estimated by linearly interpolating the first segment of the predicted capacity trajectory that crosses the failure threshold.

4.4.1. Curve-fitting metrics

Given the ground-truth capacities $\mathbf{Y} = \{y_1, \dots, y_n\}$ and predictions $\hat{\mathbf{Y}} = \{\hat{y}_1, \dots, \hat{y}_n\}$, we compute:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|, \quad (20)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}, \quad (21)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}, \quad (22)$$

where $\bar{y}$ denotes the mean of $\mathbf{Y}$.

4.4.2. RUL prediction metrics

• Absolute Error (AE):

$$\text{AE} = \left| \text{RUL}_{\text{true}} - \text{RUL}_{\text{pred}} \right|. \quad (23)$$

• Relative Error (RE):

$$\text{RE} = \frac{\left| \text{RUL}_{\text{true}} - \text{RUL}_{\text{pred}} \right|}{\text{RUL}_{\text{true}}}. \quad (24)$$

Lower values are better for MAE, RMSE, AE, and RE, while values of R^2 closer to 1 indicate a better fit.

4.5. Parameter settings

Our model was trained on a single NVIDIA RTX 4060 GPU using the Adam optimizer with a fixed input length of 64. Hyperparameters were tuned as detailed in Table 4.

Table 4
Optimal hyperparameters for OmniTIEFormer.

Hyperparameter	Search Range	CALCE	PANASONIC	GOTION
Patch length	{2, 4, 8, 16}	2	2	2
Embedding dimension	[8, 64]	8	8	24
L-HFA channel reduction	[2, 32]	2	2	2
L-HFA temporal kernel size	{3, 5, 7}	3	3	3
Encoder layers	[1, 4]	1	1	2
Encoder attention heads	[1, 8]	4	4	8
Encoder FFN factor	[1, 5]	3	3	3
Dropout rate	[0.01, 0.2]	0.1	0.1	0.01
Learning rate	[1e-4, 1e-2]	0.0001	0.001	0.001
Batch size	[64, 512]	128	128	128

5. Results and discussion

To rigorously evaluate OmniTIEFormer, we conduct extensive experiments against a range of strong baseline models. Evaluations are performed on three public datasets, namely CALCE, PANASONIC, and GOTION, which exhibit distinct degradation characteristics and collectively enable assessment under smooth monotonic fading, non-smooth transition patterns, and high-noise regimes.

5.1. Overall performance comparison

Across all three datasets, OmniTIEFormer achieves the best predictive accuracy and strong stability among the compared methods. For statistical reliability, each experiment is repeated ten times with different random seeds, while the cell-disjoint train/validation/test partition is kept fixed across runs. The random seeds affect only model initialization and stochastic training operations, and all reported metrics are averaged across these repeated runs. As summarized in Tables 5, 6, and 7, our model attains the lowest overall forecasting errors, as reflected by AMAE and ARMSE, and the highest agreement with ground truth, as indicated by AR^2 .

Importantly, these improvements are consistent across diverse degradation profiles. OmniTIEFormer performs well on smooth,

monotonic trajectories in CALCE, staircase-like transitions with regeneration in PANASONIC, and noisy, gradual decay in large-format cells in GOTION, suggesting robustness to heterogeneous trajectory shapes. A practically relevant advantage is EOL localization, where OmniTIEFormer reduces AE and RE relative to strong baselines. The following subsections provide a dataset-specific analysis of these results.

5.1.1. PANASONIC dataset

The PANASONIC dataset exhibits a staircase-like capacity degradation pattern, characterized by abrupt drops interspersed with recovery-like fluctuations (often referred to as “regeneration”) near the EOL threshold (≈ 2.1 Ah). Such non-smooth and non-monotonic behaviors pose challenges for forecasting models: approaches that aggregate patches over long windows may over-smooth sharp transitions. At the same time, highly reactive architectures can be sensitive to transient spikes. In this regime, OmniTIEFormer yields consistently strong accuracy and stability across starting points, as summarized in Table 5 and illustrated in Fig. 4. Note that *PathFormer* and *PatchFormer* are distinct baselines and are reported separately in Table 5.

Quantitative performance. As reported in Table 5 and Fig. 4(a)–(c), OmniTIEFormer attains the lowest MAE and RMSE across SP = 300, 400, and 500, while maintaining R^2 values close to 1.0, indicating faithful long-horizon trend tracking. Relative to *PatchFormer* (the strongest baseline in this setting), OmniTIEFormer reduces MAE by approximately 40% to 44% and RMSE by about 35% across starting points. More importantly for practical deployment, OmniTIEFormer improves EOL localization, reducing AE and RE by roughly 60%–74% at later starting points, where small phase errors at staircase transitions can otherwise compound into larger RUL deviations.

Qualitative trajectory analysis. Trajectory visualizations for SP = 300, 400, and 500 (Fig. 4(f)–(h)) show that OmniTIEFormer tracks the measured capacity profile with limited phase lag at step transitions and preserves the near-EOL behavior (insets). In comparison, Autoformer tends to overshoot during regeneration-like bursts, PatchFormer exhibits mild delays around sharp drops, and RUL-Mamba underfits the gradual

Table 5
Evaluation metrics of different models for the PANASONIC dataset.

Dataset	Method	SP (cycle)	TRUL (cycle)	PRUL (cycle)	MAE (Ah)	RMSE (Ah)	R^2	AE (cycle)	RE	AMAE (Ah)	ARMSE (Ah)	AR^2	AAE (cycle)	ARE
PANASONIC	ModernTCN	300	285	288.0	0.0201	0.0328	0.9717	3.0	0.0106	0.0220	0.0356	0.9427	2.13	0.0118
		400	185	187.4	0.0213	0.0350	0.9539	2.4	0.0130					
		500	85	86.0	0.0247	0.0391	0.9024	1.0	0.0119					
	Autoformer	300	285	286.8	0.0231	0.0311	0.9737	2.6	0.0092	0.0245	0.0331	0.9526	4.53	0.0218
		400	185	187.6	0.0241	0.0326	0.9585	3.8	0.0207					
		500	85	79.2	0.0263	0.0357	0.9258	7.2	0.0357					
	FEDformer	300	285	291.8	0.0257	0.0366	0.9648	6.8	0.0239	0.0274	0.0388	0.9330	7.33	0.0360
		400	185	191.8	0.0269	0.0383	0.9446	6.8	0.0370					
		500	85	92.2	0.0295	0.0415	0.8896	8.4	0.0471					
	iTrans-former	300	285	285.8	0.0162	0.0274	0.9790	1.2	0.0042	0.0180	0.0298	0.9574	1.27	0.0091
		400	185	186.2	0.0175	0.0293	0.9654	1.2	0.0065					
		500	85	86.8	0.0201	0.0325	0.9279	1.4	0.0167					
	PathFormer	300	285	299.4	0.0330	0.0399	0.9582	14.4	0.0507	0.0353	0.0436	0.9133	17.27	0.1295
		400	185	201.6	0.0346	0.0424	0.9324	16.6	0.0902					
		500	85	105.8	0.0381	0.0486	0.8493	20.8	0.2476					
	PatchFormer	300	285	283.2	0.0105	0.0192	0.9898	2.2	0.0077	0.0126	0.0213	0.9782	3.2	0.0250
		400	185	182.2	0.0120	0.0208	0.9830	3.2	0.0174					
		500	85	82.0	0.0154	0.0240	0.9616	4.2	0.0500					
	RUL-Mamba	300	285	291.0	0.0223	0.0357	0.9665	6.0	0.0211	0.0241	0.0383	0.9344	6.0	0.0417
		400	185	191.0	0.0236	0.0379	0.9460	6.0	0.0326					
		500	85	91.0	0.0263	0.0415	0.8906	6.0	0.0714					
	Ours	300	285	285.9	0.0059	0.0123	0.9959	0.9	0.0032	0.0073	0.0137	0.9909	1.0	0.0072
		400	185	186.0	0.0070	0.0133	0.9930	1.0	0.0054					
		500	85	86.3	0.0091	0.0155	0.9838	1.1	0.0131					

Note: AMAE = average MAE across all starting points; ARMSE = average RMSE across all starting points; AR^2 = average R^2 across all starting points; AAE = average AE across all starting points; ARE = average RE across all starting points.

Table 6

Evaluation metrics of different models for the CALCE dataset.

Dataset	Method	SP (cycle)	TRUL (cycle)	PRUL (cycle)	MAE (Ah)	RMSE (Ah)	R^2	AE (cycle)	RE	AMAE (Ah)	ARMSE (Ah)	AR^2	AAE (cycle)	ARE
CALCE	ModernTCN	300	341	344.0	0.0109	0.0195	0.9909	3.2	0.0094	0.0127	0.0218	0.9878	3.9	0.0204
		400	241	244.3	0.0124	0.0215	0.9888	3.3	0.0138					
		500	141	146.3	0.0147	0.0244	0.9838	5.3	0.0379					
	Autoformer	300	341	351.1	0.0216	0.0336	0.9738	10.3	0.0303	0.0246	0.0371	0.9656	11.1	0.0539
		400	241	251.8	0.0242	0.0366	0.9681	11.0	0.0458					
		500	141	152.6	0.0281	0.0410	0.9550	12.0	0.0857					
	FEDformer	300	341	355.2	0.0197	0.0301	0.9782	14.4	0.0424	0.0227	0.0332	0.9715	16.0	0.0781
		400	241	256.7	0.0224	0.0328	0.9735	15.9	0.0662					
		500	141	158.0	0.0260	0.0366	0.9628	17.6	0.1257					
	iTrans-former	300	341	347.9	0.0141	0.0229	0.9880	6.9	0.0203	0.0168	0.0256	0.9839	8.6	0.0440
		400	241	249.1	0.0166	0.0254	0.9849	8.1	0.0337					
		500	141	151.9	0.0198	0.0285	0.9778	10.9	0.0779					
	PathFormer	300	341	381.3	0.0431	0.0640	0.9060	40.3	0.1185	0.0541	0.0726	0.8695	41.9	0.2010
		400	241	282.9	0.0532	0.0713	0.8770	41.9	0.1746					
		500	141	184.4	0.0659	0.0825	0.8212	43.4	0.3100					
	PatchFormer	300	341	340.4	0.0057	0.0129	0.9962	0.8	0.0023	0.0063	0.0141	0.9951	0.9	0.0045
		400	241	240.2	0.0062	0.0140	0.9954	0.8	0.0034					
		500	141	141.1	0.0069	0.0155	0.9937	1.1	0.0078					
	RUL-Mamba	300	341	353.5	0.0164	0.0266	0.9794	12.5	0.0367	0.0194	0.0302	0.9736	12.7	0.0605
		400	241	253.5	0.0191	0.0310	0.9754	12.5	0.0520					
		500	141	154.0	0.0226	0.0331	0.9659	13.0	0.0928					
	Ours	300	341	340.7	0.0050	0.0125	0.9963	0.7	0.0006	0.0055	0.0138	0.9953	0.8	0.0020
		400	241	241.7	0.0055	0.0136	0.9956	0.7	0.0004					
		500	141	142.0	0.0061	0.0151	0.9939	1.0	0.0050					

Note: SP = starting point; AMAE = average MAE across all starting points; ARMSE = average RMSE across all starting points; AR^2 = average R^2 across all starting points; AAE = average AE across all starting points; ARE = average RE across all starting points.

Table 7

Evaluation metrics of different models for the GOTION dataset.

Dataset	Method	SP (cycle)	TRUL (cycle)	PRUL (cycle)	MAE (Ah)	RMSE (Ah)	R^2	AE (cycle)	RE	AMAE (Ah)	ARMSE (Ah)	AR^2	AAE (cycle)	ARE
GOTION	ModernTCN	450	942	950.3	0.0553	0.0750	0.9960	8.3	0.0088	0.0581	0.0784	0.9943	8.04	0.0104
		600	792	800.1	0.0570	0.0774	0.9947	8.0	0.0101					
		750	642	649.8	0.0619	0.0829	0.9921	7.8	0.0121					
	Autoformer	450	942	949.2	0.0694	0.0980	0.9930	9.9	0.0105	0.0731	0.1025	0.9901	9.33	0.0120
		600	792	798.8	0.0716	0.1011	0.9909	9.4	0.0118					
		750	642	646.6	0.0783	0.1082	0.9864	8.8	0.0137					
	FEDformer	450	942	952.5	0.0720	0.1021	0.9922	10.5	0.0111	0.0749	0.1053	0.9894	10.04	0.0129
		600	792	802.1	0.0733	0.1038	0.9902	10.1	0.0128					
		750	642	651.4	0.0794	0.1100	0.9857	9.5	0.0148					
	iTrans-former	450	942	951.7	0.0564	0.0770	0.9957	9.8	0.0104	0.0597	0.0813	0.9938	9.45	0.0122
		600	792	801.4	0.0586	0.0800	0.9944	9.5	0.0120					
		750	642	651	0.0642	0.0868	0.9913	9.0	0.0141					
	PathFormer	450	942	955.3	0.1663	0.3180	0.9276	12.8	0.0136	0.1965	0.3765	0.8613	14.90	0.0196
		600	792	808.2	0.1925	0.3704	0.8797	15.2	0.0192					
		750	642	656.4	0.2306	0.4411	0.7766	16.7	0.0260					
	PatchFormer	450	942	937.1	0.0544	0.0737	0.9961	4.8	0.0051	0.0582	0.0790	0.9941	4.86	0.0063
		600	792	787.9	0.0572	0.0780	0.9947	4.9	0.0062					
		750	642	637.2	0.0630	0.0854	0.9916	4.8	0.0075					
	RUL-Mamba	450	942	944.3	0.0602	0.0841	0.9949	2.3	0.0025	0.0636	0.0885	0.9927	2.32	0.0030
		600	792	794.3	0.0624	0.0873	0.9933	2.3	0.0029					
		750	642	644.3	0.0683	0.0943	0.9898	2.3	0.0036					
	Ours	450	942	942.9	0.0518	0.0693	0.9966	0.9	0.0010	0.0560	0.0745	0.9948	0.94	0.0012
		600	792	792.9	0.0552	0.0737	0.9952	0.9	0.0011					
		750	642	642.6	0.0609	0.0806	0.9925	1.0	0.0016					

Note: SP = starting point; AMAE = average MAE across all starting points; ARMSE = average RMSE across all starting points; AR^2 = average R^2 across all starting points; AAE = average AE across all starting points; ARE = average RE across all starting points.

inter-step drift. These qualitative behaviors are consistent with the error reductions observed in AE and RE, particularly at later starting points.

Mechanistic interpretation and key takeaway. The PANASONIC trajectories contain interacting dynamics at multiple temporal scales: a macroscopic decay trend, meso-scale transition timing around step changes,

and short-window fluctuations associated with recovery-like events and transient variability. OmniTIEFormer is designed to model these factors jointly through cross-scale interaction in TCEM, where global context helps stabilize long-horizon trend estimation, regional cues emphasize transition-related segments, and local modeling improves robustness to short-term perturbations. This interpretation is consistent

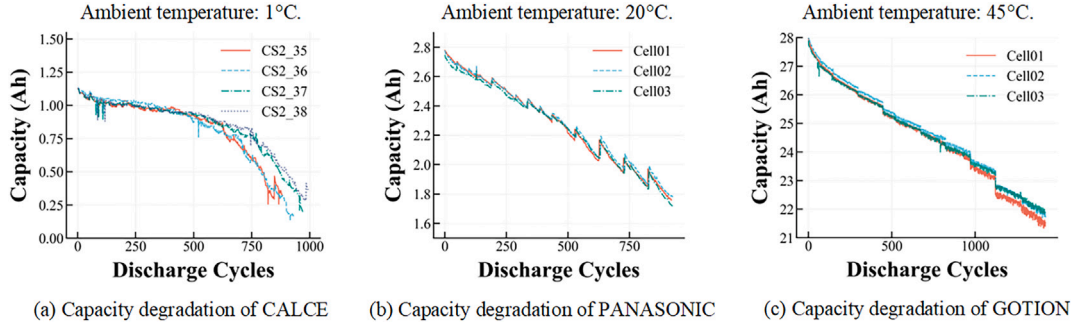


Fig. 3. Capacity degradation curves of three battery datasets: (a) CALCE, (b) PANASONIC, and (c) GOTION.

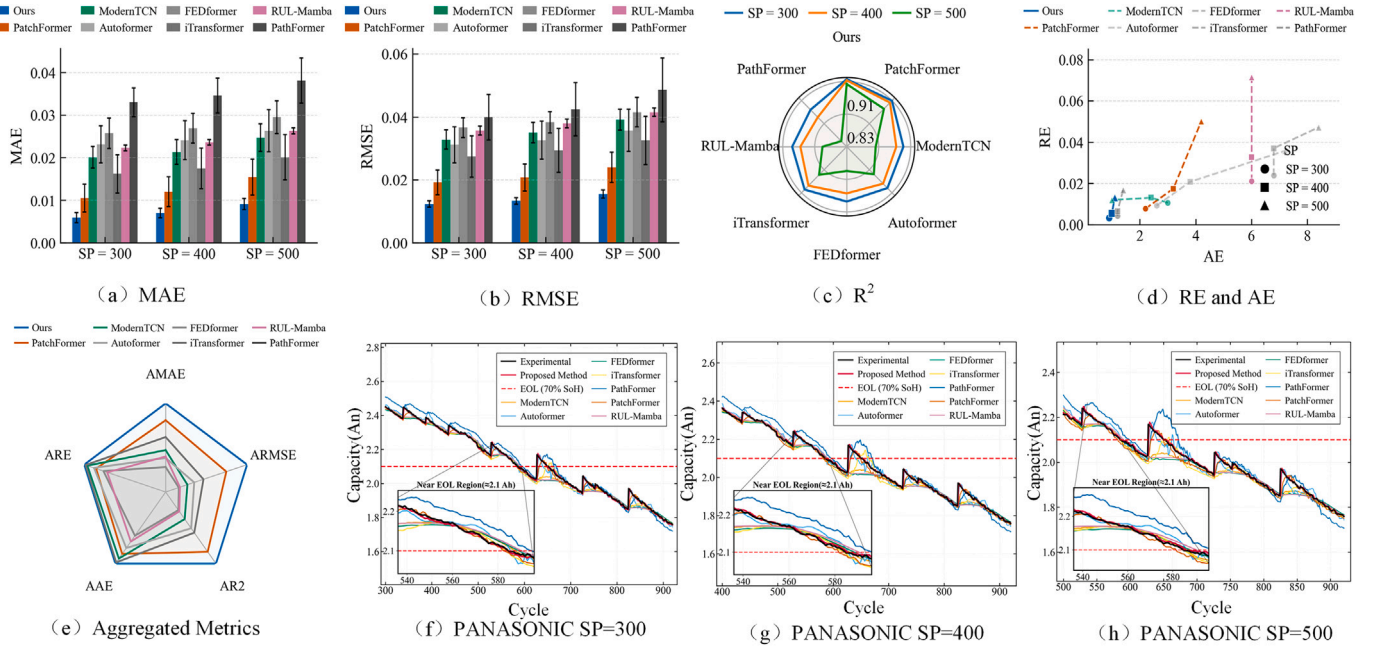


Fig. 4. Quantitative evaluation metrics and trajectory comparisons on the PANASONIC dataset. Panels (a)–(c) report MAE, RMSE, and R^2 for different starting points. Panel (d) jointly depicts the absolute error (AE) and relative error (RE) at the predicted EOL. Panel (e) presents the aggregated radar over AMAE, ARMSE, AR^2 , AAE, and ARE. Panels (f)–(h) show predicted vs. actual trajectories for starting points 300, 400, and 500.

with the combined quantitative and qualitative results: OmniTIEFormer maintains high R^2 while reducing AE/RE in the near-EOL region, suggesting that cross-scale exchange can be beneficial for non-smooth aging trajectories with regeneration-like behaviors.

5.1.2. CALCE dataset

The CALCE dataset evaluates models under a low-temperature regime with comparatively smooth, monotonic capacity fade. In this setting, OmniTIEFormer achieves strong accuracy and stability, closely tracking long-term degradation while preserving near-EOL fidelity (see Fig. 5; exact values in Table 6).

Quantitative performance. The low-temperature, monotonic fade of the CALCE dataset is comprehensively evaluated in Table 6 and Fig. 5(a)–(c). OmniTIEFormer delivers the lowest forecasting errors across all evaluated starting points. Compared with the highly competitive PatchFormer baseline, our model consistently reduces MAE by approximately 11%–12% and RMSE by roughly 3% per SP. The most notable advantage emerges in RUL prediction accuracy: OmniTIEFormer significantly bounds the RE, achieving reductions of 35.9% to 88.2%

compared to PatchFormer, depending on the starting cycle. The aggregated radar chart (Fig. 5(e)) further corroborates that OmniTIEFormer minimizes both overall curve-fitting errors (AMAE, ARMSE) and specific RUL prediction deviations (AAE, ARE) without sacrificing computational stability.

Qualitative trajectory analysis. Trajectory plots at SP = 300, 400, and 500 (Fig. 5(f)–(h)) show OmniTIEFormer aligning with the measured curve with minimal phase lag throughout mid-life and near the EOL crossing (≈ 0.77 Ah; insets). Baseline Transformers (Autoformer/FEDformer) tend to mildly over-smooth the slope and arrive late at EOL. PatchFormer shows a slight lag in the terminal region, and state-space models exhibit mild underfitting of the slow drift. These behaviors account for OmniTIEFormer's lower AE/RE in Fig. 5(d) and the favorable aggregated metrics in Fig. 5(e).

Mechanistic insight and key takeaway. Unlike the staircase profile in PANASONIC, CALCE presents a smooth, monotonic fade. Accurate forecasting hinges on anchoring the global trend while suppressing minor transient deviations near EOL. OmniTIEFormer's cross-scale exchange

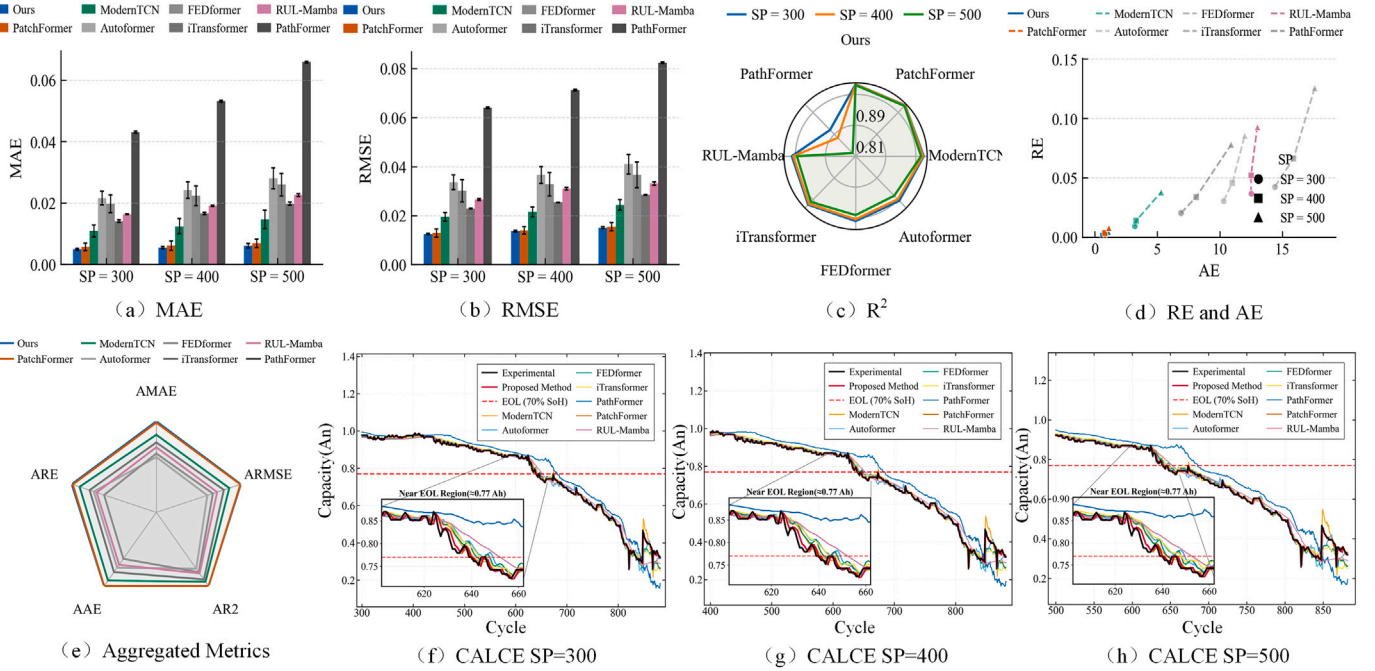


Fig. 5. Quantitative evaluation metrics and trajectory comparisons on the CALCE dataset. Panels (a)–(c) report MAE, RMSE, and R^2 for different starting points. Panel (d) shows AE and RE at the predicted EOL. Panel (e) presents the aggregated radar over AMAE, ARMSE, AR^2 , AAE, and ARE. Panels (f)–(h) show predicted vs. actual trajectories for starting points 300, 400, and 500.

(TCEM) supplies this balance: global tokens stabilize long-horizon decay, regional tokens reweight patch-wise contributions to curb horizon drift, and local tokens damp short-window noise. In this smooth, low-temperature aging regime, OmniTIEFormer delivers consistent improvements over strong baselines: roughly 11–12% lower MAE and 3% lower RMSE per SP versus PatchFormer, with substantially reduced RUL errors (RE reduced by about 36–88% across SPs). This validates cross-scale exchange for reliable trend following and precise EOL localization on CALCE.

5.1.3. GOTION dataset

The GOTION dataset represents a different operating regime from CALCE and PANASONIC. It consists of large-format 27 Ah prismatic LFP/graphite cells tested at 45 °C, with an exceptionally long cycle life, yielding substantially more training samples. As shown in Fig. 6, the capacity trajectory is characterized by a slow macroscopic decay superimposed with persistent high-frequency oscillations. These fluctuations are consistent with elevated thermal sensitivity, measurement noise, and the nonlinear electrochemical response typical of LFP chemistry.

Compared with the results on PANASONIC and CALCE, the leading methods in Fig. 6 exhibit a noticeably smaller spread in curve-fitting metrics. This observation is consistent with the statistical and physical characteristics of the GOTION dataset. First, the substantially larger training set reduces estimation variance across modern sequence models, allowing most architectures to learn the global degradation trend with comparable accuracy. Second, the degradation trajectory of large-format LFP cells evolves relatively smoothly and lacks abrupt structural transitions, which further reduces the intrinsic difficulty of learning the long-term capacity trend. Under such conditions, curve-fitting metrics alone become less discriminative among strong models.

In this regime, model differences are more clearly reflected in EOL localization and RUL estimation. Because large-format LFP batteries degrade slowly and exhibit a shallow slope near the failure threshold (set to 21.6 Ah in this study), small vertical deviations in the predicted capacity trajectory can translate into disproportionately large horizontal shifts in the inferred EOL cycle.

Quantitative performance. Consistent with this observation, several competitive baselines achieve similar curve-fitting errors. Nevertheless, OmniTIEFormer maintains the lowest MAE and RMSE while keeping R^2 values close to 1.0 across all starting points (Table 7, Fig. 6(a)–(c)). At SP = 450, for instance, MAE and RMSE decrease by 4.8% and 6.0% relative to PatchFormer.

More practically relevant differences emerge in EOL localization accuracy. OmniTIEFormer reduces AE and RE relative to RUL-Mamba by approximately 60%–62% at SP = 450 and 600, and by more than 55% at SP = 750 (e.g., AE decreases from 2.3 to 1.0). These improvements are also reflected in the aggregated radar comparison in Fig. 6(e), suggesting that the proposed architecture produces more stable long-horizon predictions under noisy industrial conditions.

Qualitative trajectory analysis. The trajectory comparisons in Fig. 6(f)–(h) provide additional insight. While most leading baselines capture the macroscopic degradation trend well, noticeable differences emerge near the EOL threshold (≈ 21.6 Ah). PatchFormer exhibits a mild phase lag during transition periods, whereas RUL-Mamba shows a gradual mid-life drift that accumulates toward the terminal region. In contrast, OmniTIEFormer suppresses high-frequency oscillations while remaining closely aligned with the measured trajectory near the EOL region, leading to more accurate RUL estimation.

Mechanistic interpretation and key takeaway. The behavior of OmniTIEFormer on the GOTION dataset is consistent with TCEM-enabled cross-scale coordination. The global pathway anchors the slow degradation trend, while the regional and local pathways adaptively mitigate transient oscillations without distorting the long-horizon phase of the trajectory. Such separation between nuisance variability and the underlying aging signal is particularly important for large-format LFP batteries, where the shallow degradation slope makes EOL localization highly sensitive to small fitting deviations. Overall, the results indicate that cross-scale information exchange contributes to robust trend tracking and improved EOL localization under high-noise industrial conditions. Within the proposed E2E lifecycle framework (Fig. 1), such stable EOL localization (e.g., an Absolute Error of only 0.9 cycles at

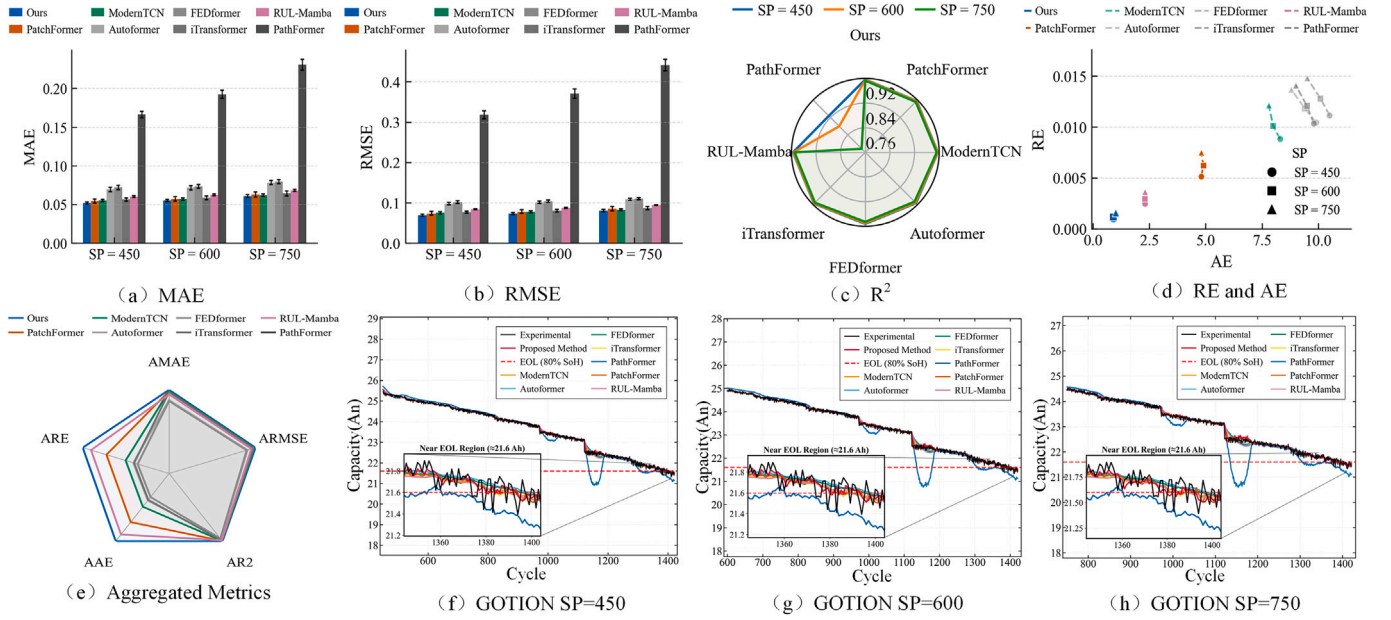


Fig. 6. Quantitative evaluation metrics and trajectory comparisons on the GOTION dataset. Panels (a)–(c) report MAE, RMSE, and R^2 for different starting points. Panel (d) jointly depicts AE and RE at the predicted EOL. Panel (e) presents the aggregated radar over AMAE, ARMSE, AR^2 , AAE, and ARE. Panels (f)–(h) show predicted vs. actual trajectories for starting points 450, 600, and 750.

SP = 450) provides the reliability required for practical lifecycle management decisions, including second-life screening and maintenance scheduling in BMS-assisted systems.

5.2. Cross-dataset, cross-chemistry, and few-shot transfer learning for robust generalization

Transfer learning for battery RUL forecasting remains highly challenging because domain shifts may occur simultaneously in battery chemistry, capacity scale, operating conditions, and degradation trajectory shape, all of which can increase the risk of negative transfer. Therefore, beyond strong within-dataset performance, a practically deployable prognostic model should be able to reuse learned representations across heterogeneous battery domains, especially when target-domain data are scarce.

Transfer settings. To evaluate this capability, we consider two representative and heterogeneous transfer pathways:

- **Cross-chemistry transfer (CALCE → PANASONIC):** transferring from low-temperature LCO cells at 1 °C with a rated capacity of 1.1 Ah to room-temperature NCA cells at 20 °C with a rated capacity of 3.03 Ah, corresponding to a shift from predominantly monotonic degradation to trajectories with more frequent non-monotonic events.
- **Cross-scale and high-noise transfer (PANASONIC → GOTION):** transferring from small-format NCA cylindrical cells to large-format 27 Ah LFP prismatic cells operated at 45 °C, corresponding to a shift from staircase-like degradation to a noisier gradual-decay regime typical of industrial applications.

Consistent gains across these two transfer settings would suggest that OmniTIEFormer learns multi-scale degradation representations that remain useful across substantially different battery domains.

Transfer protocol. We adopt a full-network fine-tuning strategy on the target domain: no layers or branches are frozen, and no grouped fine-tuning is used. All model components, including L-HFA, RCA, GCA, TCEM, and the prediction head, are jointly optimized during adaptation. Fine-tuning is performed for up to 50 epochs with early stopping based

on validation MAE (patience = 10), and the checkpoint with the lowest validation MAE is selected for evaluation. Optimization uses Adam with an initial learning rate of 1×10^{-4} and cosine annealing to $\eta_{\min} = 1 \times 10^{-5}$.

Results. As summarized in Table 8, transfer learning is particularly beneficial in data-scarce regimes. On the GOTION dataset with only 5% target-domain data, training from scratch yields an MAE of 0.0890, whereas source-domain pre-training followed by target-domain fine-tuning reduces the MAE to 0.0567. Notably, transfer learning with only 5% target data achieves performance comparable to training from scratch on the full target dataset (MAE 0.0518), highlighting the practical value of pre-training under severe data scarcity.

A similar trend is observed for CALCE → PANASONIC. With 10% target-domain data, transfer learning reduces the MAE from 0.0431 to 0.0056 under the same evaluation protocol. Across both transfer routes, the benefit of transfer learning is most pronounced at lower target-data ratios and gradually narrows as more target-domain data becomes available. In addition, pre-training improves target-domain convergence efficiency and, in many settings, reduces target-domain adaptation time relative to training from scratch.

Importantly, within the tested transfer routes and target-data ratios reported in Table 8, we do not observe negative transfer. The consistent improvements across battery chemistries, capacity scales, and degradation regimes suggest that the representations learned by OmniTIEFormer are transferable and are not overly tied to any single dataset.

Mechanistic interpretation of cross-domain generalization. A central challenge in battery transfer learning is the limited cross-domain stability of handcrafted features. In conventional feature-engineered pipelines, indicators that are informative under one charging protocol, temperature condition, or capacity scale may become unavailable, unstable, or less informative in another domain, leading to feature mismatch and weaker transfer performance. By contrast, the proposed end-to-end formulation operates directly on raw capacity sequences, encouraging the model to learn more fundamental degradation representations instead of relying on dataset-specific descriptors. This design provides a more robust basis for cross-domain adaptation, upon which TCEM can further refine representation transfer through explicit cross-scale interaction.

Table 8

Performance comparison of training from scratch (Scr.) versus transfer learning (TL) under varying data ratios.

Target	Ratio	SP	MAE (Ah)		RMSE (Ah)		Time (s)	
			Scr.	TL	Scr.	TL	Scr.	TL
PANASONIC ($\leftarrow$ CALCE)	10%	300	0.0431	0.0056	0.0476	0.0117	39.8	4.3
	20%	300	0.0405	0.0054	0.0447	0.0116	53.1	8.9
	30%	300	0.0210	0.0051	0.0347	0.0114	61.2	13.4
	50%	300	0.0139	0.0043	0.0250	0.0108	111.0	16.8
	100%	300	0.0059	0.0037	0.0123	0.0104	88.0	24.4
GOTION ($\leftarrow$ PANA.)	5%	450	0.0890	0.0567	0.1234	0.0739	6.0	4.9
	10%	450	0.0785	0.0559	0.1113	0.0729	6.5	5.7
	20%	450	0.0737	0.0537	0.1045	0.0711	18.6	6.6
	50%	450	0.0645	0.0520	0.0917	0.0685	29.9	18.0
	100%	450	0.0518	0.0504	0.0693	0.0672	71.0	33.6

Note: Scr. = training from scratch; TL = transfer learning; SP = start point; PANA. = PANASONIC. TL time denotes target-domain fine-tuning time only.

Table 9

Ablation study metrics on CALCE, PANASONIC, and GOTION datasets. L-HFA: Local Hybrid Fusion Attention, RCA: Regional Channel Attention, GCA: Global Channel Attention, TCEM: Tri-branch Cross-Exchange Module.

Dataset	Method	SP/ cycle	TRUL/ cycle	PRUL/ cycle	MAE/Ah	MAE _{std}	RMSE/Ah	RMSE _{std}	R ²	R ² _{std}	AE/cycle	AE _{std}	RE	RE _{std}
CALCE	Ours	300	341	340.7	0.0050	0.0003	0.0125	0.0002	0.9963	0.0001	0.7	0.4583	0.0006	0.0013
		400	241	241.7	0.0055	0.0004	0.0136	0.0003	0.9956	0.0002	0.7	0.4583	0.0004	0.0019
		500	141	142.0	0.0061	0.0007	0.0151	0.0004	0.9939	0.0003	1.0	0.4472	0.0050	0.0032
	w/o L-HFA	300	341	341.8	0.0062	0.0003	0.0132	0.0002	0.9960	0.0001	0.8	0.4000	0.0024	0.0012
		400	241	242.0	0.0070	0.0004	0.0145	0.0002	0.9951	0.0001	1.0	0.6325	0.0042	0.0026
		500	141	142.4	0.0081	0.0007	0.0162	0.0003	0.9931	0.0002	1.4	1.0198	0.0100	0.0073
	w/o RCA	300	341	342.0	0.0061	0.0014	0.0131	0.0011	0.9961	0.0002	1.0	0.6000	0.0029	0.0018
		400	241	242.2	0.0068	0.0017	0.0144	0.0013	0.9952	0.0002	1.2	0.6403	0.0050	0.0027
		500	141	142.4	0.0080	0.0021	0.0161	0.0014	0.9932	0.0003	1.4	0.8000	0.0035	0.0057
	w/o GCA	300	341	341.2	0.0055	0.0006	0.0128	0.0004	0.9962	0.0008	1.2	0.6000	0.0050	0.0018
		400	241	241.2	0.0061	0.0009	0.0140	0.0005	0.9954	0.0010	1.2	0.6000	0.0121	0.0025
		500	141	141.8	0.0068	0.0013	0.0155	0.0007	0.9937	0.0014	1.7	1.7916	0.0086	0.0128
	w/o TCEM	300	341	341.2	0.0064	0.0010	0.0134	0.0006	0.9959	0.0003	1.4	0.6000	0.0041	0.0018
		400	241	241.2	0.0072	0.0013	0.0146	0.0007	0.9950	0.0005	1.4	0.6000	0.0058	0.0025
		500	141	141.6	0.0082	0.0018	0.0163	0.0009	0.9930	0.0008	1.4	0.7000	0.0100	0.0050
PANASONIC	Ours	300	285	285.9	0.0059	0.0012	0.0123	0.0010	0.9959	0.0006	0.9	0.5385	0.0032	0.0019
		400	185	186.0	0.0070	0.0011	0.0133	0.0010	0.9930	0.0009	1.0	0.6708	0.0054	0.0036
		500	85	86.3	0.0091	0.0013	0.0155	0.0013	0.9838	0.0024	1.1	0.6633	0.0131	0.0078
	w/o L-HFA	300	285	283.2	0.0110	0.0025	0.0189	0.0032	0.9904	0.0029	2.0	1.4140	0.0070	0.0050
		400	185	182.8	0.0131	0.0033	0.0211	0.0039	0.9826	0.0058	2.4	1.3560	0.0130	0.0074
		500	85	82.3	0.0175	0.0050	0.0250	0.0052	0.9585	0.0154	3.3	1.5520	0.0393	0.0185
	w/o RCA	300	285	285.1	0.0067	0.0021	0.0133	0.0026	0.9951	0.0020	0.9	0.5390	0.0032	0.0019
		400	185	185.0	0.0078	0.0028	0.0144	0.0031	0.9918	0.0038	1.4	1.1140	0.0076	0.0061
		500	85	85.7	0.0100	0.0041	0.0167	0.0041	0.9812	0.0101	2.5	1.1740	0.0298	0.0215
	w/o GCA	300	285	284.2	0.0098	0.0031	0.0179	0.0043	0.9911	0.0039	1.2	0.7480	0.0042	0.0026
		400	185	183.6	0.0118	0.0041	0.0199	0.0052	0.9841	0.0074	2.2	1.1660	0.0120	0.0063
		500	85	82.6	0.0162	0.0063	0.0241	0.0068	0.9605	0.0200	3.4	2.1070	0.0405	0.0251
	w/o TCEM	300	285	287.4	0.0073	0.0010	0.0139	0.0008	0.9876	0.0005	2.4	1.8550	0.0085	0.0065
		400	185	188.4	0.0077	0.0013	0.0146	0.0009	0.9881	0.0009	3.4	2.4580	0.0185	0.0134
		500	85	89.6	0.0094	0.0018	0.0165	0.0013	0.9818	0.0026	4.6	3.4700	0.0548	0.0413
GOTION	Ours	450	942	942.9	0.0518	0.0014	0.0693	0.0024	0.9966	0.0002	0.9	0.3982	0.0010	0.0014
		600	792	793.0	0.0552	0.0013	0.0737	0.0023	0.9952	0.0003	0.9	0.4264	0.0011	0.0015
		750	642	642.6	0.0609	0.0016	0.0806	0.0026	0.9925	0.0005	1.0	0.5919	0.0016	0.0019
	w/o L-HFA	450	942	948.5	0.0555	0.0014	0.0761	0.0031	0.9958	0.0003	6.5	4.1386	0.0151	0.0044
		600	792	798.2	0.0592	0.0011	0.0819	0.0028	0.9941	0.0004	6.2	4.5880	0.0236	0.0058
		750	642	647.7	0.0672	0.0011	0.0925	0.0026	0.9900	0.0005	5.7	4.7253	0.0907	0.0074
	w/o RCA	450	942	938.3	0.0567	0.0012	0.0742	0.0011	0.9960	0.0001	3.7	3.1562	0.0085	0.0118
		600	792	788.1	0.0599	0.0012	0.0783	0.0012	0.9946	0.0001	3.9	4.4251	0.0135	0.0188
		750	642	637.3	0.0659	0.0014	0.0854	0.0015	0.9916	0.0003	4.7	4.3698	0.0339	0.0459
	w/o GCA	450	942	943.0	0.0536	0.0008	0.0707	0.0017	0.9964	0.0002	1.0	0.5191	0.0010	0.0006
		600	792	793.8	0.0553	0.0006	0.0794	0.0015	0.9952	0.0002	1.8	2.1765	0.0022	0.0027
		750	642	643.6	0.0656	0.0006	0.0875	0.0015	0.9905	0.0003	1.6	2.2429	0.0024	0.0035
	w/o TCEM	450	942	944.7	0.0565	0.0017	0.0738	0.0027	0.9961	0.0003	2.7	3.2761	0.0028	0.0035
		600	792	795.0	0.0592	0.0014	0.0774	0.0026	0.9947	0.0003	3.0	3.4676	0.0038	0.0044
		750	642	644.9	0.0650	0.0015	0.0842	0.0030	0.9918	0.0006	2.9	3.5846	0.0045	0.0056

The observed transfer gains are also consistent with TCEM's role as a dynamic cross-scale regulator rather than a static feature mixer. Across different battery domains, capacity trajectories often share a

broadly similar macroscopic fading trend, while local fluctuations, measurement noise, and meso-scale non-monotonic behaviors may vary substantially. In this setting, bidirectional cross-attention provides two

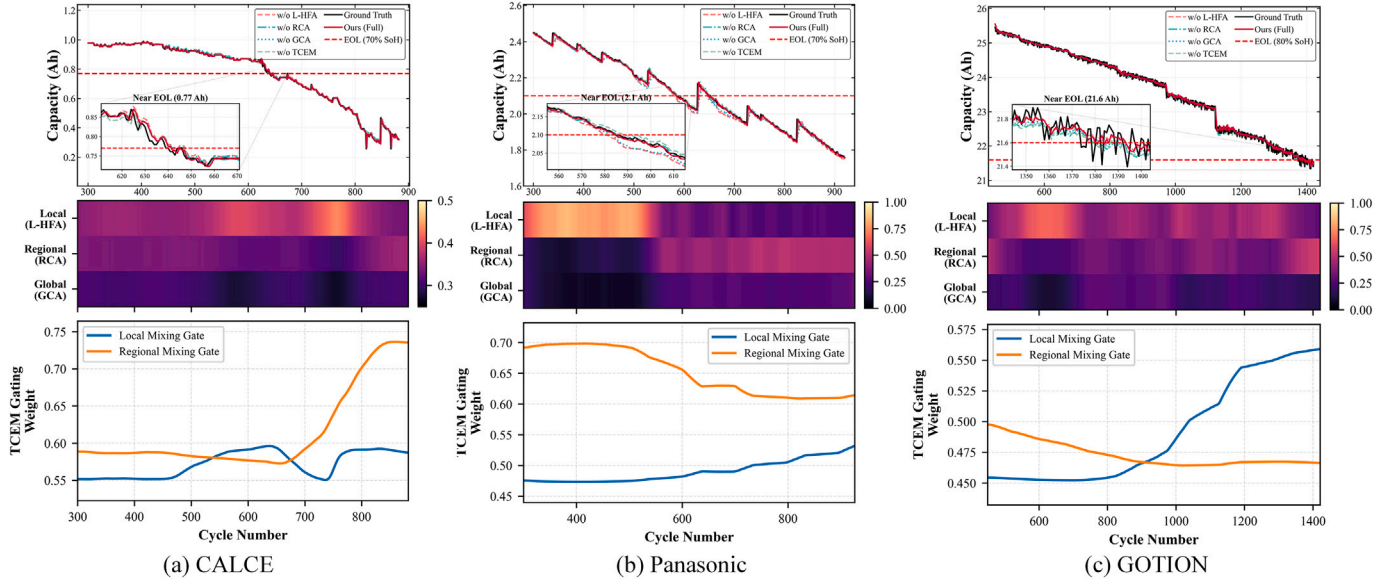


Fig. 7. Qualitative ablation on CALCE, PANASONIC, and GOTON datasets. **Top:** predicted trajectories vs. ground truth (insets: near-EOL). **Middle:** branch-wise response heatmaps for Local, Regional, and Global pathways. **Bottom:** TCEM mixing-gate weight evolution over cycles.

complementary effects: local and regional cues can adaptively modulate the global trend representation. In contrast, the global context can regularize local responses and reduce the risk that a single temporal scale dominates under distributional shifts. This interaction helps explain why OmniTIEFormer maintains strong transfer performance across heterogeneous battery systems, particularly when target-domain data are limited.

5.3. Ablation study

To evaluate the proposed multi-scale architecture and quantify the contribution of each component, namely the Local (L-HFA), Regional (RCA), Global (GCA), and TCEM hub modules, we conduct a comprehensive ablation study. We construct four variant architectures by removing one module at a time. Importantly, to isolate the effect of the *bidirectional cross-exchange*, the **w/o TCEM** variant explicitly replaces the dynamic cross-attention and gating with simple feature concatenation (i.e., a static fusion baseline). In this variant, the three branch outputs are concatenated without cross-attention or gating, while the subsequent fusion and prediction stages are kept unchanged to preserve a fair comparison. All variants are evaluated on the CALCE, PANASONIC, and GOTON datasets under an identical protocol. These ablation experiments are also used to verify the rationality of the tri-branch design itself. If one or more branches were redundant, removing them would not consistently degrade prediction performance across datasets. Instead, the observed performance drops indicate that the three branches capture complementary degradation information at different temporal scopes.

Quantitative evidence consistent with the intended module roles. Table 9 shows that the complete OmniTIEFormer consistently achieves the best overall accuracy and strong statistical robustness. In addition to lower mean errors, the full model typically exhibits smaller dispersion across runs (e.g., MAE_{std} and $RMSE_{std}$), which is particularly important for high-noise settings such as GOTON.

Notably, the degradation patterns of ablated variants are consistent with the intended roles of the three scales. On the GOTON dataset, removing the Local branch (L-HFA), which targets short-window transient fluctuations, leads to a pronounced increase in EOL localization error (e.g., AE increases from 0.9 to 6.5 at SP = 450). Replacing TCEM with the static concatenation baseline w/o TCEM also yields consistent

performance drops across datasets, suggesting that static fusion alone is insufficient to capture cross-scale interactions in non-monotonic aging trajectories. Overall, these results support the advantage of the bidirectional exchange mechanism in TCEM over a naive static fusion of branch features.

Qualitative analysis and interpretability for EOL localization. Complementary qualitative evidence is provided in Fig. 7. The trajectory comparisons (top row) show that ablated variants often exhibit prognostic lag or under-represent regeneration magnitudes. In contrast, the complete model more faithfully tracks both the long-term decay and localized recovery-like behaviors.

Moreover, the branch-wise response heatmaps (middle row) and the dynamic TCEM mixing gates (bottom row) provide additional mechanistic insight. Rather than relying on static fusion weights, the model adaptively modulates cross-scale interaction through TCEM gating. The bottom-row curves visualize the mixing-gate trajectories for the local and regional streams, showing how the amount of exchanged cross-branch information varies over the degradation process. These two curves correspond to the TCEM gating terms $g_{l \leftarrow r}$ and $g_{r \leftarrow l}$ defined in Section 3.5. This provides an interpretable view of how TCEM balances local refinement and regional coordination under different degradation regimes.

For example, the heatmaps show that the three pathways respond differently across degradation phases. Concurrently, the gate trajectories indicate that the balance between local refinement and regional coordination is dynamically adjusted over time. The global pathway remains visible in the branch-wise response map as a stabilizing context signal. In contrast, the explicit mixing gates primarily regulate the information exchanged between the local and regional streams. This phase-aware multi-scale coordination helps explain how OmniTIEFormer maintains reliable long-term prognostics while remaining robust to short-term operational variability, thereby improving EOL point localization.

5.4. Ablation on the TCEM central branch configuration

A key design choice in the TCEM is selecting a central hub for cross-scale interaction. In our default design, the exchange is anchored on the global branch, motivated by the hypothesis that sequence-level context provides the most stable reference for coordinating

Table 10
Ablation study on the TCEM central branch configuration.

Dataset	Method	SP	MAE (Ah)	RMSE (Ah)	R^2	AE	RE
CALCE	Global	300	0.0050	0.0125	0.9963	0.7	0.0006
		400	0.0055	0.0136	0.9956	0.7	0.0004
		500	0.0061	0.0151	0.9939	1.0	0.0050
	Local	300	0.0054	0.0129	0.9962	1.0	0.0029
		400	0.0059	0.0140	0.9954	1.0	0.0042
		500	0.0064	0.0155	0.9937	1.4	0.0100
	Regional	300	0.0064	0.0137	0.9956	0.7	0.0021
		400	0.0071	0.0150	0.9947	1.0	0.0042
		500	0.0080	0.0166	0.9927	1.3	0.0093
	PANASONIC	300	0.0059	0.0123	0.9959	0.9	0.0032
		400	0.0070	0.0133	0.9930	1.0	0.0054
		500	0.0091	0.0155	0.9838	1.1	0.0131
GOTION	Global	300	0.0073	0.0136	0.9947	3.2	0.0113
		400	0.0083	0.0146	0.9912	3.2	0.0174
		500	0.0101	0.0165	0.9808	4.8	0.0571
	Local	300	0.0050	0.0111	0.9967	1.0	0.0035
		400	0.0059	0.0120	0.9946	1.4	0.0076
		500	0.0074	0.0137	0.9880	1.8	0.0214
	Regional	300	0.0518	0.0693	0.9966	0.9	0.0010
		400	0.0552	0.0737	0.9952	0.9	0.0011
		500	0.0609	0.0806	0.9925	1.0	0.0016
	Local	450	0.0532	0.0700	0.9965	1.3	0.0013
		600	0.0563	0.0741	0.9952	1.3	0.0016
		750	0.0619	0.0807	0.9925	1.2	0.0019
	Regional	450	0.0529	0.0698	0.9965	2.6	0.0028
		600	0.0560	0.0739	0.9952	4.2	0.0053
		750	0.0617	0.0806	0.9925	6.3	0.0098

multi-scale features. To validate this choice, we compare the *global-centric* configuration with two alternatives: *local-centric* and *regional-centric* designs.

The results in Table 10 support our design choice. Overall, the global-centric variant yields the most consistent performance across datasets. This advantage is most evident on the GOTION dataset, where the global-centric model achieves the lowest EOL localization error (e.g., AE = 0.9 at SP = 450), compared with 1.3 and 2.6 for the local-centric and regional-centric variants, respectively. This indicates that a global contextual anchor helps regularize cross-scale exchange under noisy industrial conditions.

A notable exception is observed in the PANASONIC dataset. Specifically, the regional-centric variant yields slightly better curve-fitting performance than the global-centric design across all three starting points, as reflected by lower MAE/RMSE and higher R^2 . However, the global-centric configuration remains superior in EOL localization, achieving lower AE and RE, especially at later starting points. We attribute this divergence to the fact that the PANASONIC dataset exhibits an especially strong meso-scale staircase-like structure, as shown in Fig. 3, which may be captured more effectively when the regional branch serves as the interaction center. In contrast, the global-centric design provides a more stable sequence-level reference, which is particularly beneficial for threshold crossing localization and RUL estimation.

In summary, although different central configurations can be advantageous for specific datasets, the global-centric setting demonstrates the most consistent generalization and robustness across diverse degradation regimes, justifying its selection as the default configuration.

5.5. Sensitivity analysis of patch size

To examine the impact of the patch size on model performance, we conduct a sensitivity analysis by varying the patch length $P \in \{2, 4, 8, 16, 32\}$ across the CALCE, PANASONIC, and GOTION datasets. The results are summarized in Table 11.

Overall, $P = 2$ achieves the best overall performance across the three datasets, yielding the lowest MAE and RMSE and the highest R^2

Table 11
Sensitivity analysis of patch size across datasets.

Dataset	PATCH	SP	MAE (Ah)	RMSE (Ah)	R^2
CALCE	2	300	0.0050	0.0125	0.9963
	4	300	0.0056	0.0130	0.9961
	8	300	0.0059	0.0131	0.9961
	16	300	0.0060	0.0131	0.9959
	32	300	0.0065	0.0140	0.9955
PANASONIC	2	300	0.0059	0.0123	0.9959
	4	300	0.0061	0.0127	0.9958
	8	300	0.0066	0.0131	0.9950
	16	300	0.0076	0.0141	0.9944
	32	300	0.0108	0.0178	0.9911
GOTION	2	450	0.0518	0.0693	0.9966
	4	450	0.0535	0.0738	0.9961
	8	450	0.0529	0.0727	0.9962
	16	450	0.0547	0.0765	0.9958
	32	450	0.0568	0.0787	0.9933

for the reported evaluation settings. In general, predictive performance tends to degrade as patch size increases, although a mild non-monotonic fluctuation is observed in the GOTION dataset, where $P = 8$ slightly outperforms $P = 4$. A more noticeable degradation appears when the patch size increases to $P = 32$.

These results suggest that finer temporal granularity is generally beneficial for preserving short-term fluctuations and localized variations in battery degradation trajectories. At the same time, the optimal local aggregation scale may still vary slightly with dataset characteristics. Accordingly, $P = 2$ is adopted as the default configuration in all experiments, as it provides the most consistent overall performance.

5.6. Ablation on the TCEM temperature coefficient

In the TCEM, the cross-attention mechanism employs a temperature coefficient (τ) to control the sharpness of the attention distribution. To evaluate the benefits of introducing a learnable temperature parameter, we compare our approach with three fixed-temperature baselines:

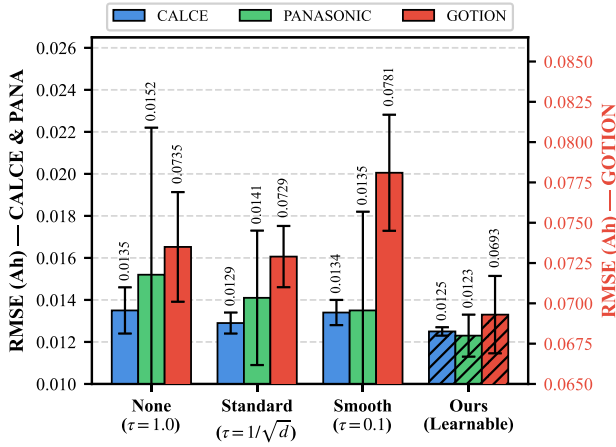


Fig. 8. Effect of the temperature coefficient (τ) on RMSE across datasets.

$\tau = 1.0$ (None), $\tau = 1/\sqrt{d}$ (standard scaled dot-product), and $\tau = 0.1$ (Smooth).

As illustrated in Fig. 8, the learnable temperature configuration consistently achieves the lowest mean RMSE across the CALCE, PANASONIC, and GOTION datasets. In addition to improving average accuracy, a key advantage of the learnable τ is the reduction in prediction variance, as reflected by the smaller error bars. For example, on the highly non-monotonic PANASONIC dataset, the fixed-temperature configurations show noticeably larger variability across different initialization seeds.

This observation suggests that allowing τ to be learned enables the model to adapt the attention sharpness to different degradation regimes. Such adaptive scaling helps suppress high-frequency fluctuations while preserving informative temporal patterns, resulting in more stable predictions across heterogeneous battery datasets.

5.7. Comparison with the classical CBAM module

We further compare the proposed L-HFA with the classical Convolutional Block Attention Module (CBAM) [49]. Both modules are evaluated under the same input settings, with CALCE and PANASONIC using $d_{\text{model}} = 8$ and GOTION using $d_{\text{model}} = 24$. The quantitative results are reported in Table 12.

From the perspective of model complexity, L-HFA introduces only a modest parameter overhead: approximately 0.103 K parameters on CALCE and PANASONIC and 0.189 K on GOTION, compared with 0.015 K for CBAM. This increase primarily stems from the lightweight MLP fusion mechanism for channel-temporal attention coupling. Notably, the absolute parameter size remains negligible relative to the overall network, thereby minimizing computational complexity.

In terms of predictive accuracy, L-HFA consistently outperforms CBAM on the CALCE dataset, achieving lower MAE and RMSE and higher R^2 . On the PANASONIC dataset, the performance gap is somewhat narrower; nevertheless, L-HFA delivers a more favorable MAE across all starting points and maintains lower overall RMSE. On the challenging GOTION dataset, characterized by higher noise and larger capacity recovery, L-HFA exhibits clear superiority: SP = 600 and SP = 750 yield more stable RMSE and R^2 , underscoring stronger robustness under noisy conditions.

In summary, L-HFA provides more comprehensive feature modeling by enabling interaction between channel and temporal attention through MLP-based fusion, rather than treating them independently as in CBAM. This joint modeling yields richer feature recalibration and more accurate, stable predictions across diverse datasets, especially in complex, noisy scenarios, while incurring only a marginal increase in the number of parameters.

Table 12

Comparison of L-HFA and CBAM across three datasets.

Dataset	Method	SP	MAE (Ah)	RMSE (Ah)	R^2	Params (K)
CALCE	L-HFA	300	0.0050	0.0125	0.9963	0.103
		400	0.0055	0.0136	0.9956	
		500	0.0061	0.0151	0.9939	
	CBAM	300	0.0055	0.0128	0.9957	0.015
		400	0.0058	0.0139	0.9943	
		500	0.0068	0.0163	0.9917	
PANASONIC	L-HFA	300	0.0059	0.0123	0.9959	0.103
		400	0.0070	0.0133	0.9930	
		500	0.0091	0.0155	0.9838	
	CBAM	300	0.0060	0.0128	0.9945	0.015
		400	0.0079	0.0142	0.9917	
		500	0.0104	0.0174	0.9815	
GOTION	L-HFA	450	0.0518	0.0693	0.9966	0.189
		600	0.0552	0.0737	0.9952	
		750	0.0609	0.0806	0.9925	
	CBAM	450	0.0547	0.0749	0.9951	0.015
		600	0.0589	0.0789	0.9929	
		750	0.0622	0.0830	0.9835	

5.8. Robustness to missing capacity data

To evaluate robustness under realistic missing-observation conditions, we use an augmented-input variant of OmniTIEFormer in this subsection only, where the original capacity input is supplemented with a binary observation mask. This setting is used exclusively for the robustness analysis and does not alter the default capacity-only formulation used in all main experiments.

In practical BMS, capacity measurements are often unavailable for every cycle due to sensor dropouts, communication interruptions, or resource constraints. As a result, recorded capacity trajectories can be discontinuous or sparsely sampled. To evaluate OmniTIEFormer under these realistic conditions, we conduct a missing-data simulation on the CALCE dataset at SP = 300.

To better reflect online estimation, we adopt a block masking strategy with causal reconstruction. Specifically, we randomly mask contiguous blocks of capacity measurements (average block length: six cycles) while varying the overall missing ratio from 0% to 40%. Missing intervals are reconstructed using forward filling (*ffill*), which avoids future information leakage but introduces step-like artifacts that can mislead sequential models. To explicitly indicate data reliability, we concatenate a binary observation mask (*obs_mask*: 1 for observed values and 0 for masked values) to the input sequence, thereby expanding the

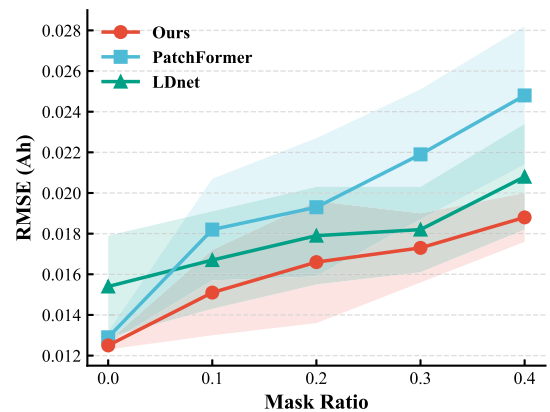


Fig. 9. Robustness to missing capacity data on the CALCE dataset (SP = 300). Block masking with forward-filling reconstruction is applied. Solid lines denote the mean RMSE, and shaded areas represent the standard deviation across multiple runs.

Table 13
Complexity and efficiency comparison across different datasets.

Dataset	Model	Params (M)	FLOPs (M)	Peak Mem. (MB)	Train Time (s)	Latency (ms/sample)		Embed. dim.	Encoder layers
						GPU	CPU		
CALCE	PatchFormer	0.098	2.48	0.95	93	10.42	8.90	16	2
	RUL-Mamba	0.075	1.65	3.61	203	8.39	11.79	48	1
	Ours	0.085	1.16	0.79	88	13.60	13.04	8	1
PANASONIC	PatchFormer	0.103	3.90	1.13	36	11.05	13.50	24	2
	RUL-Mamba	0.174	4.90	6.20	117	6.99	15.80	48	2
	Ours	0.085	1.08	0.80	32	12.62	13.60	8	1
GOTION	PatchFormer	0.109	5.46	1.36	65	10.17	9.27	32	2
	RUL-Mamba	0.145	4.65	4.23	163	7.82	15.17	32	3
	Ours	0.111	5.95	1.53	71	12.53	14.13	24	2

input dimension. We compare OmniTIEFormer with PatchFormer and the lightweight diagnostic network LDnet [50]. For a fair comparison, all compared models are provided with the same reconstructed inputs and the same binary observation mask under this augmented-input setting.

As shown in Fig. 9, RMSE increases for all models as the missing ratio grows. Nevertheless, OmniTIEFormer consistently achieves the lowest RMSE and the smallest prediction variance (shaded regions) across all missing ratios. Even with 40% of measurements missing, OmniTIEFormer remains more accurate and stable than both PatchFormer and LDnet.

Under this augmented-input setting, we attribute the observed robustness partly to TCEM's multi-scale coordination. While models that rely heavily on local continuity can be distracted by the staircase artifacts induced by *ffill*, the global and regional branches help bridge missing intervals and stabilize the macroscopic degradation trend, reducing overfitting to imputation artifacts.

5.9. Training efficiency analysis

Table 13 summarizes the computational complexity and runtime characteristics of OmniTIEFormer and competing models across datasets, including parameter count, FLOPs, peak memory usage, training time, and per-sample inference latency measured on both GPU and CPU. To maintain comparable model scales across datasets, the embedding dimension and encoder depth of OmniTIEFormer are adjusted based on sequence characteristics, resulting in similar parameter magnitudes while preserving architectural consistency.

In terms of model scale, OmniTIEFormer remains relatively compact. On the CALCE and PANASONIC datasets, the model uses 0.085M parameters, with relatively low FLOPs (1.16M and 1.08M, respectively), and a peak memory usage of approximately 0.8 MB, which is substantially lower than that of RUL-Mamba. In the experimental environment used in this study, inference latency remains stable across datasets, typically within the range of 12 to 14 ms per sample on both the GPU and the CPU. It is worth noting that inference latency does not strictly scale with FLOPs, as implementation overheads and memory-access patterns can also influence it. Overall, OmniTIEFormer offers a favorable trade-off between predictive accuracy, model footprint, and runtime cost in the reported experimental setup. While its inference latency is not uniformly lower than that of all baselines, it remains stable across datasets and is achieved with a relatively compact model size.

On the GOTION dataset, OmniTIEFormer uses 0.111M parameters, 5.95M FLOPs, and a peak memory usage of 1.53 MB. The slightly higher FLOPs compared with PatchFormer mainly arise from the longer input sequences and the additional multi-scale interactions introduced by the TCEM module. Despite this increase, the overall model scale remains comparable, and inference latency remains stable in the reported experimental environment.

In the reported experimental setup, OmniTIEFormer has the shortest training time on the PANASONIC dataset (32 s) among the compared

models, despite pronounced capacity regeneration patterns that often introduce strong non-monotonicity into the degradation trajectory. This result suggests that the proposed multi-scale architecture may help separate structured regeneration behavior from the long-term degradation trend, thereby facilitating more efficient optimization on this dataset. By contrast, the longer training times observed on CALCE (88 s) and GOTION (71 s) may reflect the combined effects of larger effective data volume and greater modeling difficulty associated with low-temperature dynamics or high-frequency noise. Overall, these results indicate that OmniTIEFormer achieves competitive training efficiency in the reported setting, while runtime may still depend on dataset characteristics, implementation details, and hardware-specific factors.

6. Conclusion

Accurately predicting the RUL of lithium-ion batteries remains a critical challenge due to complex multi-scale degradation dynamics and the limited generalization capability of existing data-driven models. In this work, we propose OmniTIEFormer, a tri-branch Transformer architecture designed to capture cross-scale temporal dependencies in battery degradation trajectories. Through its Tri-branch Cross-Exchange Module, the model enables coordinated interaction among local, regional, and global temporal representations, allowing multi-scale degradation patterns to be effectively integrated within a unified forecasting framework. Extensive evaluations under diverse aging regimes demonstrate that OmniTIEFormer consistently outperforms state-of-the-art baselines, reducing the mean absolute error by up to 44% and the EOL prediction error by up to 74%.

Cross-dataset and cross-scale transfer learning experiments further demonstrate the robustness and generalization capability of the proposed architecture. Knowledge learned from small-capacity laboratory cells can be effectively adapted to large-format industrial batteries, suggesting that the model learns degradation representations with cross-dataset utility rather than relying primarily on dataset-specific patterns. This property is particularly valuable in practical battery prognostics, where labeled datasets are often limited and operating conditions vary substantially across deployment scenarios.

Within the proposed conceptual end-to-end lifecycle management workflow, OmniTIEFormer can serve as a prognostic engine that connects algorithmic prediction with system-level battery management. Its combination of strong predictive accuracy, cross-domain adaptability, and the practical capacity-only input requirement makes it suitable for real-world battery health monitoring and prognostics. Overall, the proposed framework contributes to more reliable battery health assessment and lifecycle management by enabling accurate RUL estimation, thereby supporting safety-aware operational decisions and second-life evaluation in BMS-assisted systems.

Future work will focus on three main directions. First, we plan to incorporate physics-informed constraints into the multi-scale architecture

to improve interpretability and physical consistency. Second, integrating multimodal electrochemical signals (e.g., voltage and temperature) could further enhance prognostic fidelity under highly dynamic operating conditions. Finally, while this study establishes the algorithmic core and the closed-loop E2E workflow, advancing toward hardware-in-the-loop deployment remains a critical next step. Integrating the proposed framework into real-time Battery Management Systems will enable validation of its capability to support hardware-level scheduling for second-life utilization, maintenance planning, and battery recycling.

CRedit authorship contribution statement

Xuan Yi: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jianmao Xiao:** Writing – review & editing, Supervision, Project administration, Funding acquisition. **Gang Lei:** Supervision, Funding acquisition. **Xin Hu:** Visualization, Validation. **Musheng Wu:** Supervision, Funding acquisition. **Bo Xu:** Formal analysis. **Chuying Ouyang:** Writing – review & editing, Formal analysis.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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